



# **RTL Design Sherpa**

## **RLB Top Micro-Architecture Specification 1.0**

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# 1 Rlb Top Mas Index

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## 2 RLB Top - Overview

### 2.1 What This Block Is

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`rlb_top` is the integration level of the Retro Legacy Blocks subsystem. It takes one APB4 slave port and presents, behind it, a complete set of PC-compatible legacy peripherals: two interrupt controllers, an IOAPIC, two timers, a real-time clock, an SMBus controller, a GPIO controller, a UART, and a power-management block.

It is not a peripheral. It contains no registers of its own, implements no protocol, and every address it answers belongs to something it instantiates. What it does contribute is three things that exist nowhere else in the subsystem:

1. **An address map.** A 1-to-10 APB crossbar splits one slave port into ten 4KB windows from `BASE_ADDR`, and answers an unmapped address with `PSLVERR` rather than hanging.
2. **An interrupt fabric.** Each block's interrupt output is routed internally onto its conventional legacy line and delivered to both 8259s and to the IOAPIC, so a board does not have to wire ten interrupt pins back in.
3. **A cascade.** The two 8259 instances are cross-connected as a PC/AT master and slave pair, with the vector and acknowledge path between them.

### 2.2 Why It Is a Separate Specification

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Each of the nine other books in this directory specifies one block. None of them can specify the above, because none of them can see it: a block knows its own interrupt output pin, not which controller that pin reaches, and certainly not what happens when two blocks assert at once.

Before this book existed, the routing table and the bring-up order lived only in the testbench under `dv/tbclasses/rlb_top/`. That is a workable place to verify behaviour and a poor place to look it up.

## 2.3 Block Inventory

Ten windows, all assigned. There is no reserved window.

| Window | Address range             | Block                    | Instance    | Specification                  |
|--------|---------------------------|--------------------------|-------------|--------------------------------|
| 0      | 0xFEC0_0000 - 0xFEC0_0FFF | HPET                     | u_hpet      | <a href="#">hpet_mas</a>       |
| 1      | 0xFEC0_1000 - 0xFEC0_1FFF | 8259 PIC (master)        | u_pic       | <a href="#">pic_8259_mas</a>   |
| 2      | 0xFEC0_2000 - 0xFEC0_2FFF | 8254 PIT                 | u_pit       | <a href="#">pit_8254_mas</a>   |
| 3      | 0xFEC0_3000 - 0xFEC0_3FFF | RTC                      | u_rtc       | <a href="#">rtc_mas</a>        |
| 4      | 0xFEC0_4000 - 0xFEC0_4FFF | SMBus                    | u_smbus     | <a href="#">smbus_mas</a>      |
| 5      | 0xFEC0_5000 - 0xFEC0_5FFF | PM/ACPI                  | u_pm_acpi   | <a href="#">pm_acpi_mas</a>    |
| 6      | 0xFEC0_6000 - 0xFEC0_6FFF | IOAPIC                   | u_ioapic    | <a href="#">ioapic_mas</a>     |
| 7      | 0xFEC0_7000 - 0xFEC0_7FFF | GPIO                     | u_gpio      | <a href="#">gpio_mas</a>       |
| 8      | 0xFEC0_8000 - 0xFEC0_8FFF | UART 16550               | u_uart      | <a href="#">uart_16550_mas</a> |
| 9      | 0xFEC0_9000 - 0xFEC0_9FFF | 8259 PIC (cascade slave) | u_pic_slave | <a href="#">pic_8259_mas</a>   |

Two further instances carry no address window:

| Instance           | Module           | Contribution                                                           |
|--------------------|------------------|------------------------------------------------------------------------|
| u_apbx_xbar        | apbx_xbar_1to10  | The address decode. Generated, not hand-written.                       |
| u_ioapic_boot_intx | ioapic_boot_intx | Turns “the IOAPIC is not delivering this pin” into a legacy PIC input. |

Twelve instances in total.

## 2.4 Features

---

- Single APB4 slave entry point, 32-bit address and data, with PSTRB and PPROT carried through to every block
- Ten 4KB windows on a clean power-of-two decode: the slave index is PADDR[15:12]
- Unmapped addresses complete with PSLVERR instead of stalling the bus
- Interrupt fabric covering six sourcing blocks, presented on the conventional legacy IRQ lines
- Both 8259s cross-connected as a cascaded pair, with master IR2 protected
- A single aggregated `rlb_irq_out` for a system that wants one interrupt input rather than ten
- Boot-interrupt rerouting available and disabled by default
- External `pic_irq_in` and `ioapic_irq_in` retained as inputs and OR-ed with the internal fabric, so an existing board keeps its interrupt path

## 2.5 Deliberate Scope Boundaries

---

These are decisions, not defects. Each is recorded here because each one looks like a bug to somebody reading the RTL for the first time.

**IRQ2 is never driven by the fabric.** In a PC/AT pair, master IR2 belongs to the cascade: it carries the slave controller's INT, not a device. `rlb_top` masks IR2 off every other source and forces it from the slave, so an interrupt routed to IRQ2 reaches nothing. This is also why IRQ8 exists as a line at all.

**The general HPET timers are not routed.** `hpet_timer_irq` has no conventional legacy line, which is the same reason the HPET's `TIMER_INT_ROUTE_CAP` field reads 0. Those timers reach `rlb_irq_out` and their own output pins. Only the two legacy-replacement outputs, `hpet_legacy_irq0` and `hpet_legacy_irq8`, enter the fabric.

**ioapic\_msi\_emit is not instantiated.** Software can program `cfg_msi_addr` and `cfg_msi_data` through the IOAPIC's IOWIN, but nothing in this subsystem consumes them. The pins are connected and deliberately left open rather than omitted, because an omitted pin is a PINMISSING warning that hides the gap.

**Boot-interrupt rerouting has a default map, not a board map.** The pin to legacy-input mapping is a board and chipset decision. The default is the published identity mapping for pins 0-7 with no reroute above that, and rerouting is disabled at reset.

**Every block is instantiated with `CDC_ENABLE(0)`.** The per-block clock and reset ports are present on `rlb_top`'s boundary but unused in this configuration. See [Clocks and Reset](#).

## 2.6 Applications

- A drop-in legacy peripheral subsystem for a soft-core SoC on FPGA
- A target for firmware and driver development against PC-compatible peripherals without the hardware
- A worked example of an integration level: generated address decode, an interrupt fabric with convention-versus-choice documented, and a cascade

## 2.7 Related Documents

- [Architecture](#) - how the pieces fit together
- [Top-Level Interface](#) - the integration contract
- [Initialization](#) - bring-up order
- [Window Map](#) - the address map in detail

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# 3 RLB Top - Architecture

## 3.1 Hierarchy

|                      |                  |                          |               |
|----------------------|------------------|--------------------------|---------------|
| rlb_top              |                  |                          |               |
| — u_apbx_xbar        | apbx_xbar_1to10  | one master, ten slaves   |               |
| (GENERATED)          |                  |                          |               |
| — u_hpet             | apb4_hpet        | window 0                 |               |
| — u_pic              | apb4_pic_8259    | window 1                 | master of the |
| pair                 |                  |                          |               |
| — u_pic_slave        | apb4_pic_8259    | window 9                 | slave of the  |
| pair                 |                  |                          |               |
| — u_pit              | apb4_pit_8254    | window 2                 |               |
| — u_rtc              | apb4_rtc         | window 3                 |               |
| — u_smbus            | apb4_smbus       | window 4                 |               |
| — u_pm_acpi          | apb4_pm_acpi     | window 5                 |               |
| — u_ioapic           | apb4_ioapic      | window 6                 |               |
| — u_gpio             | apb4_gpio        | window 7                 |               |
| — u_uart             | apb4_uart_16550  | window 8                 |               |
| — u_ioapic_boot_intx | ioapic_boot_intx | no window; combinational |               |
| companion            |                  |                          |               |

Everything else in the module is wiring and two pieces of combinational logic: the interrupt fabric and the cascade cross-connect.

## 3.2 The Address Path

One APB4 slave port enters the crossbar. The crossbar decodes PADDR[15:12] into a slave index and forwards the transaction, including PSTRB and PPROT, to that slave's port.

**The crossbar is generated.** It comes from the shared crossbar generator, configured as one master, ten slaves, base 0xFEC00000, 32-bit address and data, 4KB per slave. Do not hand-edit `apbx_xbar_1to10.sv`. The generator owns the decode-miss path, where an unmapped address completes with PSLVERR instead of hanging, and a hand-rolled copy is exactly how that behaviour gets lost - it already happened once to the crossbar this one replaced.

**The crossbar routes the full address; each block takes the low 12 bits.** Every peripheral instantiation slices PADDR[11:0] off its window's channel. A block therefore sees offsets from 0x000 within its own window and needs no knowledge of BASE\_ADDR. This is the one detail most likely to trip up a reader comparing the top-level port width (32 bits) against a block's (12 bits).

| Property                | Value                          |
|-------------------------|--------------------------------|
| Base address            | BASE_ADDR, default 0xFEC0_0000 |
| Window size             | 4KB (0x1000)                   |
| Windows                 | 10, indices 0-9, none reserved |
| Slave index             | PADDR[15:12]                   |
| Address seen by a block | PADDR[11:0] of its own window  |
| Unmapped address        | completes with PSLVERR         |

## 3.3 The Interrupt Fabric

Each block emits its interrupt on its own output pin, and those pins remain on the boundary. The fabric additionally collects them into an internal 16-bit vector, `w_fabric_irq`, on the conventional legacy line numbers, and feeds that vector to three destinations.

### 3.3.1 Source to line

| Source signal                              | Line | Basis                        | Reaches                          |
|--------------------------------------------|------|------------------------------|----------------------------------|
| pit_timer_irq[0]<br>OR<br>hpet_legacy_irq0 | IRQ0 | convention<br>(system timer) | master 8259 IR0,<br>IOAPIC pin 0 |
| uart_irq                                   | IRQ4 | convention<br>(COM1)         | master 8259 IR4,<br>IOAPIC pin 4 |
| rtc_alarm_irq OR                           | IRQ8 | convention (RTC              | slave 8259 IR0,                  |

| Source signal                            | Line  | Basis                                  | Reaches                              |
|------------------------------------------|-------|----------------------------------------|--------------------------------------|
| rtc_second_irq<br>OR<br>hpet_legacy_irq8 |       | alarm)                                 | IOAPIC pin 8                         |
| pm_interrupt                             | IRQ9  | convention (ACPI)                      | slave 8259 IR1,<br>IOAPIC pin 9      |
| smb_interrupt                            | IRQ10 | <b>choice</b> , parameter<br>IRQ_SMBUS | slave 8259 IR2,<br>IOAPIC pin 10     |
| gpio_irq                                 | IRQ11 | <b>choice</b> , parameter<br>IRQ_GPIO  | slave 8259 IR3,<br>IOAPIC pin 11     |
| hpet_timer_irq[*]<br>]                   | none  | no conventional<br>line exists         | rlb_irq_out and<br>its own pins only |
| -                                        | IRQ2  | reserved for the<br>cascade            | never driven by<br>the fabric        |

Convention versus choice is a real distinction here, and it is reflected in the RTL. IRQ0, IRQ4, IRQ8 and IRQ9 come from the published typical-assignment table reproduced in the IOAPIC specification, so they are fixed as localparam inside the module: making them overridable would invite silently diverging from what every driver already assumes. SMBus and GPIO have no traditional assignment - the table offers only “Available” - so those two are this subsystem’s choice and are exposed as module parameters an integrator can move.

Two sources share IRQ0 and three share IRQ8. That is intentional and safe: HPET legacy-replacement mode exists precisely to replace the 8254 tick and the RTC periodic interrupt, and the HPET core suppresses timers 0 and 1 on hpet\_timer\_irq while that mode is active, so the same event cannot arrive twice.

### 3.3.2 The three destinations

**Master 8259.** The master sees

```
w_master_pic_irq = ((pic_irq_in[7:0] | w_boot_intx_pic_irq |
w_fabric_irq[7:0]) & 8'hFB)
                  | {5'b0, w_spic_int, 2'b0}
```

Note the order: the external input, the boot-interrupt reroute and the fabric are OR-ed together first, then & 8'hFB clears bit 2, and only then is bit 2 driven from the slave’s INT. The mask is what makes the cascade trustworthy - without it an external IRQ2, or a boot-interrupt reroute onto pin 2, could impersonate the slave controller. Every other master level passes through unchanged.

**Slave 8259.** The slave's `irq_in` is the inline expression `pic_irq_in[15:8] | w_fabric_irq[15:8]`. It is not a named signal, which is worth knowing when probing: with `pic_irq_in` held at zero, the slave's IR line for a given IRQ is `w_fabric_irq[irq]`.

**IOAPIC.** `w_ioapic_irq = ioapic_irq_in | w_fabric_irq`, zero-extended to the IOAPIC's pin count. The fabric defines 16 legacy lines; the IOAPIC has `IOAPIC_NUM_IRQS` pins, default 24, and the upper pins are for additional devices this subsystem does not source.

### 3.3.3 OR-ed in, never substituted

`pic_irq_in` and `ioapic_irq_in` remain inputs and are OR-ed with the fabric. This is deliberate: an integrator keeps the external interrupt path, and every test that drove those pins directly still works. Replacing them would have been a silent change to the port contract.

`pic_int_out` is **not** a fabric source. Feeding the controller's own output back into its inputs is a combinational loop. It appears only in the aggregate below, which drives nothing internally.

## 3.4 The Cascade Cross-Connect

The two 8259 instances are wired as a PC/AT pair.

| Direction       | Signal                     | Purpose                                                        |
|-----------------|----------------------------|----------------------------------------------------------------|
| slave to master | <code>w_spic_int</code>    | the slave's INT, forced onto master IR2                        |
| master to slave | <code>w_pic_cas_ack</code> | the master's cascade acknowledge                               |
| slave to master | <code>w_spic_vector</code> | the slave's vector, returned by the master in place of its own |

The master's `cas_ack_in` ties low, because nothing acknowledges the master from above. The invariant this creates is that master IR2 always equals `w_spic_int` - not that IR2 never moves. For any IRQ8-15 source, IR2 *should* rise; that is the cascade working.

## 3.5 The Aggregated Output

`rlb_irq_out` is a single line asserted while any block interrupt is asserted:

```
rlb_irq_out = |{hpet_timer_irq, hpet_legacy_irq0, hpet_legacy_irq8,
               pic_int_out, pit_timer_irq,
               rtc_alarm_irq, rtc_second_irq, smb_interrupt,
               pm_interrupt, gpio_irq, uart_irq}
```

It exists for a system that wants one interrupt input rather than ten pins. It is a pure OR, it includes `pic_int_out` because that is what a single-line consumer wants to see, and it takes no

part in the routing above - which is precisely why including the controller's own output is safe here.

## 3.6 Boot-Interrupt Rerouting

---

The IOAPIC exports which pins it is *not* delivering. `u_ioapic_boot_intx` turns that into legacy PIC inputs, so a masked IOAPIC pin can still reach the 8259.

The map is the published identity mapping - IOAPIC pin  $n$  to legacy input  $n$  for  $n$  in 0 to 7 - narrowed to the eight inputs the master controller has, with a no-reroute code for pins 8 and above. Pin 2 is a dead entry in that table: it still names legacy input 2, but master IR2 is masked off the external inputs and driven from the slave, so a reroute onto pin 2 reaches nothing. It is left in the map so the table still reads as the identity mapping; an integrator who needs pin 2 rerouted picks a different legacy input.

Rerouting is **safe by default**: the enable bit resets to 0, so `w_boot_intx_pic_irq` is all zeros and the master's OR is exactly `pic_irq_in` until software opts in.

## 3.7 Related Documents

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- [Overview](#)
- [Clocks and Reset](#)
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- [Initialization](#)

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# 4 RLB Top - Clocks and Reset

## 4.1 The Short Version

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There is one functional clock domain in this configuration: `pc_lk`. Every other clock port exists on the boundary and is unused.

That needs stating plainly, because the port list suggests otherwise.

## 4.2 Clock and Reset Ports

| Clock      | Reset         | Intended domain     | Used in this configuration        |
|------------|---------------|---------------------|-----------------------------------|
| pclk       | presetn       | APB, primary        | <b>Yes</b> - everything runs here |
| hpet_clk   | hpet_resetn   | HPET counter        | No                                |
| pit_clk    | pit_resetn    | PIT counting clock  | No                                |
| rtc_clk    | rtc_resetn    | RTC time base       | No                                |
| smbus_clk  | smbus_resetn  | SMBus bit clock     | No                                |
| pm_clk     | pm_resetn     | PM/ACPI             | No                                |
| ioapic_clk | ioapic_resetn | IOAPIC              | No                                |
| gpio_clk   | gpio_rstn     | GPIO sampling       | No                                |
| uart_clk   | uart_rstn     | UART baud reference | No                                |

Note the naming inconsistency in the last two: GPIO and UART use `gpio_rstn` and `uart_rstn`, not `_resetn`. That is the blocks' own port naming, carried through.

## 4.3 Why the Extra Clocks Are Unused

`rlb_top` instantiates every CDC-capable block with `CDC_ENABLE(0)`:

| Instance               | Parameterisation           |
|------------------------|----------------------------|
| <code>u_hpet</code>    | <code>CDC_ENABLE(0)</code> |
| <code>u_pit</code>     | <code>CDC_ENABLE(0)</code> |
| <code>u_smbus</code>   | <code>CDC_ENABLE(0)</code> |
| <code>u_pm_acpi</code> | <code>CDC_ENABLE(0)</code> |
| <code>u_ioapic</code>  | <code>CDC_ENABLE(0)</code> |
| <code>u_gpio</code>    | <code>CDC_ENABLE(0)</code> |
| <code>u_uart</code>    | <code>CDC_ENABLE(0)</code> |

With `CDC_ENABLE=0` a block uses the plain APB slave and ignores its second clock and reset entirely. The ports remain on `rlb_top` so the boundary does not change if a future configuration enables CDC on one or more blocks, but today nothing downstream of them is clocked.

**What an integrator should do:** tie every unused clock to `pclk` and every unused reset to `presetn`. Leaving them unconnected or tied to a constant is legal in this configuration but becomes a real bug the moment any block is rebuilt with `CDC_ENABLE=1`, and the failure then looks like a dead peripheral rather than a missing clock.

The RTC is the case worth flagging: a real system usually wants its time base on a separate always-on 32.768 kHz domain, which in this subsystem means instantiating `apb4_rtc` with CDC enabled rather than relying on `rtc_clk` as wired here.

## 4.4 Reset Behaviour

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- **Active low, asynchronous**, throughout. There is no positive-polarity reset anywhere in this subsystem.
- Reset is applied through the repository's reset macros from `reset_defs.svh`, which `rlb_top` includes. A hand-written `always_ff @(posedge clk or negedge rst_n)` is not used and would be rejected.
- `rlb_top` itself holds no state. It has no `always_ff` of its own: the interrupt fabric is a single `always_comb`, and the cascade and aggregate are continuous assignments. All reset behaviour belongs to the instantiated blocks.

### 4.4.1 Reset state of the interrupt fabric

Because the fabric is purely combinational, its reset state is whatever its sources are, and at reset those are all deasserted. Two consequences matter for bring-up:

- **`w_boot_intx_pic_irq` is all zeros after reset**, because the IOAPIC's boot-interrupt enable resets to 0. The master's interrupt input is therefore exactly `pic_irq_in` until software enables rerouting.
- **IOAPIC redirection entries reset masked**. A source can reach the IOAPIC's pin and produce no delivery at all, which looks identical to a broken route. Unmasking the relevant entry is a required bring-up step, not an optimisation. See [Initialization](#).

### 4.4.2 Resetting to clear an interrupt

The 8259 in edge-triggered mode latches INT high until it is acknowledged. The integration test suite resets the whole subsystem between phases rather than acknowledging, because the acknowledge and EOI paths have their own known limitations at block level. That is a verification convenience and not a recommendation for firmware; software should acknowledge properly. It is mentioned here because it explains why the testbench re-programs both controllers and re-arms the IOAPIC after every such phase - a full reset drops all of that configuration.

## 4.5 Related Documents

- [Architecture](#)
- [Top-Level Interface](#)
- Repository reset and clocking rules: `vault/handbook/design/reset-and-clocking.md`

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## 5 RLB Top - Acronyms and Terminology

### 5.1 Acronyms

| Term   | Expansion                                  | In this subsystem                             |
|--------|--------------------------------------------|-----------------------------------------------|
| ACPI   | Advanced Configuration and Power Interface | The power-management block on window 5        |
| APB    | Advanced Peripheral Bus                    | The bus every block speaks; APB4 here         |
| APB4   | AMBA 4 APB                                 | The revision in use, with PSTRB and PPROT     |
| CDC    | Clock Domain Crossing                      | Parameter CDC_ENABLE, 0 in this configuration |
| EOI    | End Of Interrupt                           | Acknowledge written back to a controller      |
| GPE    | General Purpose Event                      | ACPI wake and status events, pm_gpe_events    |
| GPIO   | General Purpose Input/Output               | The controller on window 7                    |
| HPET   | High Precision Event Timer                 | The timer on window 0                         |
| ICW    | Initialization Command Word                | 8259 configuration writes, ICW1-ICW4          |
| IOAPIC | I/O Advanced                               | The controller on                             |

| Term  | Expansion                                   | In this subsystem                            |
|-------|---------------------------------------------|----------------------------------------------|
|       | Programmable Interrupt Controller           | window 6                                     |
| IR    | Interrupt Request line                      | An individual input of an 8259, IR0-IR7      |
| IRQ   | Interrupt Request                           | A legacy line number, IRQ0-IRQ15             |
| MSI   | Message Signalled Interrupt                 | Configurable in the IOAPIC, not emitted here |
| OCW   | Operation Command Word                      | 8259 runtime writes, OCW1-OCW3               |
| PIC   | Programmable Interrupt Controller           | The 8259 pair, windows 1 and 9               |
| PIT   | Programmable Interval Timer                 | The 8254 on window 2                         |
| PM    | Power Management                            | Used interchangeably with ACPI here          |
| RLB   | Retro Legacy Blocks                         | This subsystem                               |
| RTC   | Real-Time Clock                             | The block on window 3                        |
| RTE   | Redirection Table Entry                     | An IOAPIC per-pin configuration entry        |
| SMBus | System Management Bus                       | The controller on window 4                   |
| UART  | Universal Asynchronous Receiver/Transmitter | The 16550 on window 8                        |

## 5.2 Terminology

**Window.** One of the ten 4KB address regions behind the crossbar. Window  $n$  starts at  $\text{BASE\_ADDR} + n * 0x1000$ . The window number equals the crossbar's slave index, which equals  $\text{PADDR}[15:12]$ .

**Fabric.** The internal interrupt routing in `rlb_top`: the `w_fabric_irq` vector and the logic that builds it and distributes it. Not a bus.

**Line.** A legacy IRQ number, 0 to 15. A line is a position in the fabric vector, and separately an input on one of the two controllers.

**Convention versus choice.** A line assignment is *convention* if it comes from the published typical-assignment table, and a *choice* if this subsystem picked it because no traditional assignment exists. Convention assignments are fixed in the RTL; choices are module parameters.

**Master and slave.** The two 8259 instances. The master is on window 1 and drives `pic_int_out`; the slave is on window 9 and reaches the CPU only through the master's IR2. Unrelated to the APB sense of master and slave.

**Cascade.** The connection between the two controllers. Master IR2 carries the slave's INT, the master's acknowledge reaches the slave, and the slave's vector is returned in place of the master's.

**Boot interrupt.** A source whose IOAPIC pin is masked, rerouted to the legacy controller so it is not lost during boot before the IOAPIC is configured.

**Legacy replacement.** An HPET mode in which HPET timers 0 and 1 take over the 8254 tick and the RTC periodic interrupt, appearing on `hpet_legacy_irq0` and `hpet_legacy_irq8` while being suppressed on `hpet_timer_irq`.

## 5.3 Related Documents

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- [Overview](#)
- [References](#)

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# 6 RLB Top - References

## 6.1 External Specifications

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Every peripheral in this subsystem implements a published specification. Copies of the specifications used are kept in the component's `References/` directory.

| Block    | Specification                                           |
|----------|---------------------------------------------------------|
| HPET     | High Precision Event Timer specification, revision 1.0a |
| 8259 PIC | 8259A programmable interrupt controller datasheet       |
| 8254 PIT | 82C54 interval timer datasheet                          |
| RTC      | MC146818 real-time clock plus                           |

| Block      | Specification                                            |
|------------|----------------------------------------------------------|
|            | RAM datasheet                                            |
| SMBus      | System Management Bus specification, version 3.3         |
| PM/ACPI    | Advanced Configuration and Power Interface specification |
| IOAPIC     | I/O advanced programmable interrupt controller datasheet |
| UART 16550 | 16550 asynchronous communications element datasheet      |
| Bus        | AMBA APB protocol specification, APB4                    |

The interrupt line assignments this subsystem treats as convention - IRQ0 for the system timer, IRQ4 for COM1, IRQ8 for the RTC alarm and IRQ9 for ACPI - come from the typical-assignment table reproduced in [ioapic\\_mas ch05](#). That table is the authority cited in the RTL, and it is also the evidence that IRQ10 and IRQ11 have no traditional assignment, which is why SMBus and GPIO are parameters rather than fixed.

## 6.2 In-Repository References

### 6.2.1 This subsystem

| Item                   | Location                                              |
|------------------------|-------------------------------------------------------|
| Integration RTL        | <a href="#">rtl/rlb_top/rlb_top.sv</a>                |
| Integration filelist   | <a href="#">rtl/rlb_top/filelists/rlb_top.f</a>       |
| Generated crossbar     | <a href="#">rtl/apbx_xbar/apbx_xbar_1to10.sv</a>      |
| Test runner            | <a href="#">dv/tests/test_rlb_top.py</a>              |
| Testbench classes      | <a href="#">dv/tbclasses/rlb_top/</a>                 |
| Subsystem requirements | <a href="#">PRD.md</a>                                |
| Subsystem overview     | <a href="#">README.md</a>                             |
| Area facts and traps   | <a href="#">CLAUDE.md</a>                             |
| FPGA integration guide | <a href="#">docs/RLB_FPGA_IMPLEMENTATION_GUIDE.md</a> |

| Item                            | Location                                           |
|---------------------------------|----------------------------------------------------|
| Implementation status           | <a href="#">docs/IMPLEMENTATION_STATUS.md</a>      |
| Driver development reading list | <a href="#">docs/LegacyBlocksAndDriverGuide.md</a> |

### 6.2.2 Per-block specifications

| Block      | Specification                  |
|------------|--------------------------------|
| HPET       | <a href="#">hpet_mas</a>       |
| 8259 PIC   | <a href="#">pic_8259_mas</a>   |
| 8254 PIT   | <a href="#">pit_8254_mas</a>   |
| RTC        | <a href="#">rtc_mas</a>        |
| SMBus      | <a href="#">smbus_mas</a>      |
| PM/ACPI    | <a href="#">pm_acpi_mas</a>    |
| IOAPIC     | <a href="#">ioapic_mas</a>     |
| GPIO       | <a href="#">gpio_mas</a>       |
| UART 16550 | <a href="#">uart_16550_mas</a> |

### 6.2.3 Repository practice

The methods this subsystem follows are recorded in the handbook rather than restated here:

| Topic                    | Note                                                              |
|--------------------------|-------------------------------------------------------------------|
| Generated RTL discipline | <a href="#">vault/handbook/design/generated-rtl-discipline.md</a> |
| Reset and clocking       | <a href="#">vault/handbook/design/reset-and-clocking.md</a>       |
| Filelists                | <a href="#">vault/handbook/design/filelists.md</a>                |
| Running regressions      | <a href="#">vault/handbook/dv/running-regressions.md</a>          |
| Register map links       | <a href="#">vault/handbook/authoring/register-map-links.md</a>    |

## 6.3 Related Documents

- [Overview](#)
- [Acronyms](#)

## 7 RLB Top - Block Hierarchy Overview

### 7.1 The Twelve Instances

`rlb_top` contains no logic blocks of its own. It instantiates twelve modules and wires them together, with two pieces of combinational glue: the interrupt fabric and the cascade cross-connect.

| Instance                 | Module                       | Window | Parameters passed                                                                                                     |
|--------------------------|------------------------------|--------|-----------------------------------------------------------------------------------------------------------------------|
| <code>u_apbx_xbar</code> | <code>apbx_xbar_1to10</code> | -      | <code>ADDR_WIDTH</code> ,<br><code>DATA_WIDTH</code> ,<br><code>STRB_WIDTH</code> ,<br><code>BASE_ADDR</code>         |
| <code>u_hpet</code>      | <code>apb4_hpet</code>       | 0      | <code>VENDOR_ID</code> ,<br><code>REVISION_ID</code> ,<br><code>NUM_TIMERS</code> ,<br><code>CDC_ENABLE(0)</code>     |
| <code>u_pic</code>       | <code>apb4_pic_8259</code>   | 1      | <code>none</code>                                                                                                     |
| <code>u_pit</code>       | <code>apb4_pit_8254</code>   | 2      | <code>NUM_COUNTERS</code> ,<br><code>CDC_ENABLE(0)</code>                                                             |
| <code>u_rtc</code>       | <code>apb4_rtc</code>        | 3      | <code>none</code>                                                                                                     |
| <code>u_smbus</code>     | <code>apb4_smbus</code>      | 4      | <code>FIFO_DEPTH</code> ,<br><code>CDC_ENABLE(0)</code>                                                               |
| <code>u_pm_acpi</code>   | <code>apb4_pm_acpi</code>    | 5      | <code>CDC_ENABLE(0)</code>                                                                                            |
| <code>u_ioapic</code>    | <code>apb4_ioapic</code>     | 6      | <code>NUM_IRQS</code> ,<br><code>CDC_ENABLE(0)</code>                                                                 |
| <code>u_gpio</code>      | <code>apb4_gpio</code>       | 7      | <code>GPIO_WIDTH</code> ,<br><code>SYNC_STAGES</code> ,<br><code>CDC_ENABLE(0)</code> ,<br><code>SKID_DEPTH(2)</code> |
| <code>u_uart</code>      | <code>apb4_uart_16550</code> | 8      | <code>FIFO_DEPTH</code> ,<br><code>SYNC_STAGES</code> ,<br><code>CDC_ENABLE(0)</code> ,<br><code>SKID_DEPTH(2)</code> |

| Instance            | Module           | Window | Parameters passed                                    |
|---------------------|------------------|--------|------------------------------------------------------|
| u_pic_slave         | apb4_pic_8259    | 9      | none                                                 |
| u_ioapic_boot_in_tx | ioapic_boot_intx | -      | NUM_IRQS,<br>NUM_PIC(8),<br>PIC_IDX_W(4),<br>PIC_MAP |

## 7.2 u\_apbx\_xbar - the address decode

A generated 1-to-10 APB crossbar. It is produced by the shared crossbar generator, configured as one master, ten slaves, base 0xFEC00000, 32-bit address and data, and 4KB per slave.

**Do not hand-edit it.** The generator owns the decode-miss path - an unmapped address completes with PSLVERR rather than hanging the bus - and the hand-rolled crossbar this one replaced had already lost that behaviour. It is regenerated, not patched.

The crossbar forwards the full 32-bit address. Each peripheral instantiation slices PADDR[11:0] from its own channel, so a block sees offsets from 0x000 within its window.

## 7.3 The two 8259 instances

The same module appears twice, and the difference is entirely in the wiring.

|               | u_pic (master, window 1)               | u_pic_slave (slave, window 9)                          |
|---------------|----------------------------------------|--------------------------------------------------------|
| irq_in        | w_master_pic_irq (a named signal)      | pic_irq_in[15:8] \  <br>w_fabric_irq[15:8]<br>(inline) |
| int_out       | pic_int_out, a module output           | w_spic_int, internal only                              |
| cas_vector    | w_spic_vector, from the slave          | 8'h00, tied off                                        |
| cas_ack_in    | 1'b0 - nothing acknowledges the master | w_pic_cas_ack, from the master                         |
| cas_ack       | w_pic_cas_ack, to the slave            | unconnected                                            |
| inta_vector_o | unconnected                            | w_spic_vector, to the master                           |

Two details are worth carrying away. First, the slave's `irq_in` is an **inline expression, not a named signal**, so there is nothing to probe by name; with `pic_irq_in` held at zero the slave's IR

line for a given IRQ is exactly `w_fabric_irq[irq]`. Second, the master's IR2 is not OR-ed with anything - it is masked off every other source and then forced from the slave.

## 7.4 u\_ioapic - and what it is not wired to

The IOAPIC receives `w_ioapic_irq`, which is the external `ioapic_irq_in` OR-ed with the zero-extended fabric vector. Its delivery handshake, EOI input and retry input are all on the module boundary.

Four of its configuration outputs are **connected and deliberately left open**:

| Pin                           | Why it is open                                                                               |
|-------------------------------|----------------------------------------------------------------------------------------------|
| <code>cfg_msi_addr</code>     | <code>ioapic_msi_emit</code> is not instantiated here, so nothing consumes MSI configuration |
| <code>cfg_msi_data</code>     | as above                                                                                     |
| <code>cfg_mask_vec</code>     | consumed by <code>u_ioapic_boot_intx</code>                                                  |
| <code>cfg_boot_intx_en</code> | consumed by <code>u_ioapic_boot_intx</code>                                                  |

The first two are the interesting case. Software can program them through IOWIN, and nothing will happen. They are wired to an empty connection rather than omitted from the instantiation because an omitted pin produces a PINMISSING warning, which is how a gap like this becomes invisible.

## 7.5 u\_ioapic\_boot\_intx - the boot-interrupt companion

A small combinational module. It takes the raw IOAPIC pin inputs, the IOAPIC's mask vector and its boot-interrupt enable, and produces legacy PIC inputs for the pins the IOAPIC is not delivering.

Its map, `BOOT_INTX_PIC_MAP`, is built as a localparam in `rlb_top`: the identity mapping for pins 0 through 7, and a no-reroute code for every pin above that. Field 0 is the least significant, so the identity entries are written last in the concatenation.

Pin 2 is a **dead entry**. The map still names legacy input 2, but master IR2 is masked off the external inputs and driven from the slave controller, so a reroute onto pin 2 reaches nothing. It is kept in the map so the table still reads as the published identity mapping; an integrator who needs pin 2 rerouted must choose a different legacy input.

Because the IOAPIC's boot-interrupt enable resets to 0, this module's output is all zeros after reset and the master's interrupt input is exactly `pic_irq_in` until software opts in.

## 7.6 The glue

Neither of these is a module, and both are worth naming because they are where the integration behaviour lives.

**The interrupt fabric** is one `always_comb` block that builds `w_fabric_irq[15:0]` from the blocks' interrupt outputs, plus three continuous assignments that distribute it: to the IOAPIC, to the master controller with the IR2 mask, and into the `rlb_irq_out` aggregate.

**The cascade cross-connect** is three wires between the two controllers, listed in the table above.

`rlb_top` has no `always_ff` of its own. It holds no state.

## 7.7 Related Documents

- [Architecture](#) - the fabric and cascade in detail
- [Top-Level Interface](#) - the port list
- [Window Map](#)

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# 8 RLB Top - Top-Level Interface

## 8.1 Overview

`rlb_top` is the integration contract for the whole subsystem: one APB4 slave port, the peripherals' external pins, and one aggregated interrupt output. This chapter is the pin list, the parameters, and the address decode rules an integrator has to honour.

## 8.2 Parameters

| Parameter               | Type             | Default | Description                                                                    |
|-------------------------|------------------|---------|--------------------------------------------------------------------------------|
| <code>ADDR_WIDTH</code> | <code>int</code> | 32      | APB address width. The subsystem decodes a full system address, not a windowed |

| Parameter        | Type                      | Default      | Description                                                          |
|------------------|---------------------------|--------------|----------------------------------------------------------------------|
|                  |                           |              | one.                                                                 |
| DATA_WIDTH       | int                       | 32           | APB data width.                                                      |
| STRB_WIDTH       | int                       | DATA_WIDTH/8 | Write-strobe width. Derived; do not override independently.          |
| BASE_ADDR        | logic<br>[ADDR_WIDTH-1:0] | 32'hFEC00000 | Base of the ten-window region.                                       |
| HPET_VENDOR_ID   | int                       | 1            | Driven into the HPET capability register.                            |
| HPET_REVISION    | int                       | 1            | Driven into the HPET capability register.                            |
| HPET_NUM_TIMERS  | int                       | 2            | HPET comparator count. Sets the width of hpet_timer_irq.             |
| PIT_NUM_COUNTERS | int                       | 3            | 8254 counter count. Sets the width of pit_gate_in and pit_timer_irq. |
| IOAPIC_NUM_IRQS  | int                       | 24           | IOAPIC pin count. Sets the width of ioapic_irq_in.                   |
| GPIO_WIDTH       | int                       | 32           | GPIO pin count.                                                      |
| GPIO_SYNC_STAGES | int                       | 2            | GPIO input synchroniser depth.                                       |
| UART_FIFO_DEPTH  | int                       | 16           | UART FIFO depth.                                                     |
| UART_SYNC_STAGES | int                       | 2            | UART input synchroniser                                              |

| Parameter        | Type | Default | Description                                                        |
|------------------|------|---------|--------------------------------------------------------------------|
| SMBUS_FIFO_DEPTH | int  | 32      | depth.<br>SMBus FIFO depth.                                        |
| IRQ_SMBUS        | int  | 10      | <b>Choice, not convention.</b> The legacy line SMBus is routed to. |
| IRQ_GPIO         | int  | 11      | <b>Choice, not convention.</b> The legacy line GPIO is routed to.  |

### 8.2.1 Why only two interrupt lines are parameters

IRQ0, IRQ4, IRQ8 and IRQ9 are fixed as `localparam` inside the module because they come from the published typical-assignment table: the system timer, COM1, the RTC alarm and ACPI respectively. Exposing them would invite silently diverging from what every driver already assumes.

SMBus and GPIO have no traditional assignment - the table lists IRQ10 and IRQ11 as available - so those two are this subsystem's choice and are parameters an integrator can move.

## 8.3 Ports

### 8.3.1 Clock and Reset

| Port                     | Dir | Width  | Notes                                                       |
|--------------------------|-----|--------|-------------------------------------------------------------|
| pclk                     | in  | 1      | APB clock. The only functional clock in this configuration. |
| presetn                  | in  | 1      | APB reset, active low, asynchronous.                        |
| hpet_clk,<br>hpet_resetn | in  | 1 each | Unused: u_hpet is CDC_ENABLE(0). Tie to pclk / presetn.     |

| Port                          | Dir | Width  | Notes                                                       |
|-------------------------------|-----|--------|-------------------------------------------------------------|
| pit_clk,<br>pit_resetrn       | in  | 1 each | Unused:<br>CDC_ENABLE(0).<br>Tie to pclk /<br>presetrn.     |
| rtc_clk,<br>rtc_resetrn       | in  | 1 each | Unused in this<br>configuration. Tie<br>to pclk / presetrn. |
| smbus_clk,<br>smbus_resetrn   | in  | 1 each | Unused:<br>CDC_ENABLE(0).<br>Tie to pclk /<br>presetrn.     |
| pm_clk, pm_resetrn            | in  | 1 each | Unused:<br>CDC_ENABLE(0).<br>Tie to pclk /<br>presetrn.     |
| ioapic_clk,<br>ioapic_resetrn | in  | 1 each | Unused:<br>CDC_ENABLE(0).<br>Tie to pclk /<br>presetrn.     |
| gpio_clk,<br>gpio_rstn        | in  | 1 each | Unused:<br>CDC_ENABLE(0).<br>Note _rstn, not<br>_resetrn.   |
| uart_clk,<br>uart_rstn        | in  | 1 each | Unused:<br>CDC_ENABLE(0).<br>Note _rstn, not<br>_resetrn.   |

See [Clocks and Reset](#) for why the secondary clocks are present but unused, and what to do about it.

### 8.3.2 APB4 Slave Interface

The single entry point to the subsystem.

| Port       | Dir | Width | Notes |
|------------|-----|-------|-------|
| s_apb_PSEL | in  | 1     |       |

| Port          | Dir | Width      | Notes                                             |
|---------------|-----|------------|---------------------------------------------------|
| s_apb_PENABLE | in  | 1          |                                                   |
| s_apb_PADDR   | in  | ADDR_WIDTH | <b>Full system address</b> , not a window offset. |
| s_apb_PWRITE  | in  | 1          |                                                   |
| s_apb_PWDATA  | in  | DATA_WIDTH |                                                   |
| s_apb_PSTRB   | in  | STRB_WIDTH | Carried through to the selected block.            |
| s_apb_PPROT   | in  | 3          | Carried through to the selected block.            |
| s_apb_PRDATA  | out | DATA_WIDTH |                                                   |
| s_apb_PSLVERR | out | 1          | Also the decode-miss response.                    |
| s_apb_PREADY  | out | 1          |                                                   |

### 8.3.3 Peripheral External Interfaces

| Block    | Inputs                                                                                    | Outputs                                                                       |
|----------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| HPET     | -                                                                                         | hpet_timer_irq[HPET_NUM_TIMERS-1:0],<br>hpet_legacy_irq0,<br>hpet_legacy_irq8 |
| 8259 PIC | pic_irq_in[15:0]                                                                          | pic_int_out                                                                   |
| 8254 PIT | pit_gate_in[PIT_NUM_COUNTERS-1:0]                                                         | pit_timer_irq[PIT_NUM_COUNTERS-1:0]                                           |
| RTC      | -                                                                                         | rtc_alarm_irq,<br>rtc_second_irq                                              |
| SMBus    | smb_scl_i, smb_sda_i                                                                      | smb_scl_o, smb_scl_t,<br>smb_sda_o, smb_sda_t,<br>smb_interrupt               |
| PM/ACPI  | pm_gpe_events[31:0],<br>pm_gpe1_events[31:0],<br>pm_power_button_n,<br>pm_sleep_button_n, | pm_clock_gate_en[31:0],<br>pm_power_domain_en[7:0], pm_sys_reset_req,         |

| Block      | Inputs                                                                                                                            | Outputs                                                                                                                                          |
|------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| IOAPIC     | pm_rtc_alarm,<br>pm_ext_wake_n,<br>pm_wdt_reset_n,<br>pm_ext_reset_n,<br>pm_power_domain_ack[7:0]                                 | pm_periph_reset_req,<br>pm_interrupt                                                                                                             |
|            | ioapic_irq_in[IOAPIC_NUM_IRQS-1:0],<br>ioapic_irq_out_ready,<br>ioapic_irq_out_retry,<br>ioapic_eoi_in,<br>ioapic_eoi_vector[7:0] | ioapic_irq_out_valid,<br>ioapic_irq_out_vector[7:0],<br>ioapic_irq_out_dest[7:0],<br>ioapic_irq_out_deliv_mode[2:0],<br>ioapic_irq_out_dest_mode |
| GPIO       | gpio_in[GPIO_WIDTH-1:0]                                                                                                           | gpio_out[GPIO_WIDTH-1:0],<br>gpio_oe[GPIO_WIDTH-1:0], gpio_irq                                                                                   |
| UART 16550 | uart_rx, uart_cts_n,<br>uart_dsr_n, uart_rn_n,<br>uart_dcd_n                                                                      | uart_tx, uart_dtr_n,<br>uart_rts_n,<br>uart_rxrdy_n,<br>uart_txrdy_n,<br>uart_out1_n,<br>uart_out2_n, uart_irq                                   |
| Subsystem  | -                                                                                                                                 | rlb_irq_out                                                                                                                                      |

#### 8.3.4 Ports that need care

| Port                 | What to know                                                                                                         |
|----------------------|----------------------------------------------------------------------------------------------------------------------|
| pic_irq_in[2]        | <b>Ignored.</b> Master IR2 is the cascade input, forced from the slave controller. Driving this bit reaches nothing. |
| pic_irq_in[15:8]     | Goes to the <b>slave</b> controller, not the master.                                                                 |
| ioapic_irq_out_retry | Tie low if the receiver always accepts. It is the IOAPIC's half of delegated lowest-priority delivery.               |

| Port                           | What to know                                                                                                          |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| pm_power_domain_ack[7:0]       | Tie high unless the rail sequencer is enabled with an acknowledge requirement.                                        |
| pm_wdt_reset_n, pm_ext_reset_n | Tie high (inactive) if the system has no watchdog or reset button. They are reported in the PM reset-status register. |
| pm_gpe1_events[31:0]           | Tie to zero when the system has only one GPE bank.                                                                    |
| uart_rxdy_n, uart_txdy_n       | DMA handshake pins. Leave unconnected in a programmed-I/O system.                                                     |
| hpet_timer_irq                 | Not routed into the interrupt fabric. Wire it yourself if you need it delivered.                                      |
| rlb_irq_out                    | A pure OR of every block interrupt including pic_int_out. Drives nothing internally.                                  |

## 8.4 Address Decode Contract

| Property               | Value                                                      |
|------------------------|------------------------------------------------------------|
| Base                   | BASE_ADDR, default 0xFEC0_0000                             |
| Window size            | 0x1000 (4KB)                                               |
| Window count           | 10, indices 0-9                                            |
| Slave index            | PADDR[15:12]                                               |
| Offset seen by a block | PADDR[11:0]                                                |
| Reserved windows       | none - all ten are assigned                                |
| Unmapped address       | completes normally with PSLVERR asserted; it does not hang |

The decode-miss behaviour is a property of the generated crossbar, and it is the reason the crossbar must be regenerated rather than hand-edited.

## 8.5 Integration Checklist

---

1. Drive `pclk` and `presetn`. Tie every secondary clock to `pclk` and every secondary reset to `presetn` unless you have rebuilt a block with `CDC_ENABLE=1`.
2. Connect the APB4 slave port, passing the **full** system address.
3. Decide whether you want the internal interrupt fabric, the external pins, or both. Both is the default and costs nothing: they are OR-ed.
4. If you are not driving external legacy interrupts, tie `pic_irq_in` and `ioapic_irq_in` to zero. The fabric then supplies everything.
5. Tie off the PM pins listed above according to what your system actually has.
6. Choose `IRQ_SMBUS` and `IRQ_GPIO` if the defaults clash with your platform.
7. Decide how the IOAPIC's delivery handshake is consumed, and tie `ioapic_irq_out_retry` low if your receiver always accepts.
8. Use `rlb_irq_out` if you want a single interrupt input; otherwise use `pic_int_out` and the IOAPIC delivery interface.

## 8.6 Related Documents

---

- [Architecture](#)
- [Clocks and Reset](#)
- [Initialization](#)
- [Window Map](#)

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---

# 9 RLB Top - Initialization and Programming Guide

## 9.1 Scope

---

This chapter is the **subsystem** bring-up: the order in which the pieces have to be configured, and the parts that only exist at the integration level - the interrupt controllers, the IOAPIC redirection table, and the cascade. Programming a block's own function (setting a timer period, configuring a UART's baud rate) belongs to that block's specification.

The sequences below are the ones the integration test suite executes, which is why they are known to work rather than merely plausible.

## 9.2 Address Helper

---

Every register access in this chapter goes through a window. One helper keeps the rest readable:

```
#define RLB_BASE      0xFEC00000u
#define RLB_WINDOW    0x1000u

#define WIN_HPET      0
#define WIN_PIC        1      /* master 8259 */
#define WIN_PIT        2
#define WIN_RTC        3
#define WIN_SMBUS      4
#define WIN_PM         5
#define WIN_IOAPIC     6
#define WIN_GPIO       7
#define WIN_UART       8
#define WIN_PIC_SLAVE  9      /* slave 8259 */

static inline volatile uint32_t *rlb_reg(unsigned window, unsigned
offset)
{
    return (volatile uint32_t *) (RLB_BASE + window * RLB_WINDOW +
offset);
}
```

## 9.3 Bring-Up Order

---

| Step | What                                                | Why here                                                                      |
|------|-----------------------------------------------------|-------------------------------------------------------------------------------|
| 1    | Release reset, confirm the subsystem answers        | Nothing below is meaningful if decode is broken                               |
| 2    | Decide the interrupt topology                       | Determines whether step 3 is the single or cascade sequence                   |
| 3    | Configure the 8259 (single <b>or</b> cascaded pair) | The controllers must be initialised before any source is enabled              |
| 4    | Configure IOAPIC redirection entries                | Entries reset <b>masked</b> ; an unconfigured entry silently delivers nothing |

| Step | What                         | Why here                                        |
|------|------------------------------|-------------------------------------------------|
| 5    | Enable the peripheral blocks | Only now can a source assert without being lost |
| 6    | Verify                       | Read back status rather than assuming           |

The ordering constraint that actually matters is steps 3 and 4 before step 5. A block enabled before its controller is configured asserts into a controller that is still in its initialisation state, and in edge-triggered mode the result is a latched interrupt nobody is ready to service.

## 9.4 Step 1: Confirm the Subsystem Answers

Each window has a read-safe probe register. Reading all ten is the cheapest possible confirmation that the crossbar and every block are alive.

```
/* Read-safe probe per window. */
static const struct { unsigned window, offset; const char *name; }
probes[] = {
    { WIN_HPET,      0x000, "hpet"      }, /* HPET_ID, read-only
identity */
    { WIN_PIC,       0x000, "pic"        }, /* PIC_CONFIG */
    { WIN_PIT,       0x000, "pit"        }, /* PIT_CONFIG */
    { WIN_RTC,       0x000, "rtc"        }, /* RTC_CONFIG */
    { WIN_SMBUS,     0x000, "smbus"      }, /* SMBUS_CONTROL */
    { WIN_PM,        0x000, "pm"         }, /* ACPI_CONTROL */
    { WIN_IOAPIC,    0x000, "ioapic"     }, /* IOREGSEL */
    { WIN_GPIO,      0x000, "gpio"       }, /* GPIO_CONTROL */
    { WIN_UART,      0x020, "uart"       }, /* UART_SCR -- see the
warning */
    { WIN_PIC_SLAVE, 0x000, "pic_slave" }, /* PIC_CONFIG on the slave
*/
};
```

**Do not probe the UART at offset 0x000.** That is the receive buffer, and reading it pops a byte off the RX FIFO. Use the scratch register at 0x020. This is a real trap: a “harmless” identity read that destroys received data.

An unmapped address is also a useful probe in the other direction: it must complete with PSLVERR asserted rather than hang. If it hangs, the crossbar is not the generated one.

## 9.5 Step 2: Choose the Interrupt Topology

| Topology    | Use when                             | Configure             |
|-------------|--------------------------------------|-----------------------|
| Single 8259 | Nothing you care about is on IRQ8-15 | Master only, SNGL set |

| Topology      | Use when                                                           | Configure                    |
|---------------|--------------------------------------------------------------------|------------------------------|
| Cascaded pair | Anything on IRQ8-15 - which includes the RTC, ACPI, SMBus and GPIO | Both controllers, SNGL clear |
| IOAPIC        | You want vectored delivery with per-pin destination control        | Redirection entries, step 4  |

The fabric feeds all of these simultaneously. Choosing one does not disable the others.

Note which sources land where: **IRQ0 and IRQ4 reach the master; IRQ8, IRQ9, IRQ10 and IRQ11 reach the slave.** With four of the six sourcing blocks on the slave controller, the single-controller topology is rarely what you want.

## 9.6 Step 3a: Single-Controller Configuration

Master only, in single mode. This is enough to get INT asserting.

```
#define PIC_CONFIG 0x000
#define PIC_ICW1 0x004
#define PIC_ICW2 0x008
#define PIC_ICW3 0x00C
#define PIC_ICW4 0x010
#define PIC_OCW1 0x014
#define PIC_STATUS 0x018

int pic_init_single(uint8_t vector_base) /* vector_base e.g. 0x20 */
{
    *rlb_reg(WIN_PIC, PIC_CONFIG) = 0x1; /* enable, NOT
init_mode */
    *rlb_reg(WIN_PIC, PIC_ICW1) = 0x10 | 0x02 | 0x01; /* marker |
SNGL | IC4 */
    *rlb_reg(WIN_PIC, PIC_ICW2) = vector_base;
    /* No ICW3: SNGL is set, so the controller does not expect one. */
    *rlb_reg(WIN_PIC, PIC_ICW4) = 0x01; /* 8086 mode */
    *rlb_reg(WIN_PIC, PIC_OCW1) = 0x00; /* unmask all */

    return (*rlb_reg(WIN_PIC, PIC_STATUS) & 1) != 0; /* init
complete */
}
```

**PIC\_CONFIG must set the enable bit without init\_mode.** Setting `init_mode` here sends the initialisation state machine back to its idle state once the sequence completes, and the controller ends up unconfigured. This is the single most likely way to get a controller that accepts every write and then does nothing.

ICW1 here is 0x13: the marker bit, SNGL, and IC4. Level-triggered mode (LTIM) is left clear, so the controller is edge-triggered - which matters when clearing an interrupt, because INT latches until acknowledged.

## 9.7 Step 3b: Cascaded-Pair Configuration

---

Both controllers, as a master and slave pair. Anything arriving on IRQ8-15 requires this.

```
int pic_init_cascade(uint8_t master_base, uint8_t slave_base) /* e.g.
0x20, 0x28 */
{
    /* ---- MASTER, window 1 */
    *rlb_reg(WIN_PIC, PIC_CONFIG) = 0x1;
    *rlb_reg(WIN_PIC, PIC_ICW1)   = 0x10 | 0x01; /* marker | IC4,
SNGL CLEAR */
    *rlb_reg(WIN_PIC, PIC_ICW2)   = master_base;
    *rlb_reg(WIN_PIC, PIC_ICW3)   = 0x04;      /* a slave is
attached on IR2 */
    *rlb_reg(WIN_PIC, PIC_ICW4)   = 0x01;
    *rlb_reg(WIN_PIC, PIC_OCW1)   = 0x00;      /* unmask all, IR2
included */

    /* ---- SLAVE, window 9 */
    *rlb_reg(WIN_PIC_SLAVE, PIC_CONFIG) = 0x1;
    *rlb_reg(WIN_PIC_SLAVE, PIC_ICW1)   = 0x10 | 0x01;
    *rlb_reg(WIN_PIC_SLAVE, PIC_ICW2)   = slave_base;
    *rlb_reg(WIN_PIC_SLAVE, PIC_ICW3)   = 0x02; /* I hang off master
IR2 */
    *rlb_reg(WIN_PIC_SLAVE, PIC_ICW4)   = 0x01;
    *rlb_reg(WIN_PIC_SLAVE, PIC_OCW1)   = 0x00;

    return (*rlb_reg(WIN_PIC, PIC_STATUS) & 1)
        && (*rlb_reg(WIN_PIC_SLAVE, PIC_STATUS) & 1);
}
```

Four things must all be true for a slave interrupt to reach the CPU, and omitting any one of them looks exactly like a broken interrupt fabric:

1. The master is in cascade mode - SNGL **clear** in ICW1
2. The master's ICW3 marks IR2 as carrying a slave (0x04)
3. The slave's ICW3 states which master line it hangs off (0x02)
4. Master IR2 is unmasked in OCW1

**With SNGL clear, ICW3 is mandatory.** The initialisation state machine waits for it. Skipping it leaves the controller stuck mid-sequence.

Note that `master_base` and `slave_base` differ by 8 (0x20 and 0x28): the slave's eight lines get their own vector block.

## 9.8 Step 4: IOAPIC Redirection Entries

---

The IOAPIC is programmed indirectly: write the register index to `IOREGSEL`, then the value to `IOWIN`.

```
#define IOAPIC_IOREGSEL 0x000
#define IOAPIC_IOWIN    0x010
#define IOAPIC_BOOTINTX /* see ioapic_mas ch05 for the offset */

static void ioapic_write(unsigned selector, uint32_t value)
{
    *rlb_reg(WIN_IOAPIC, IOAPIC_IOREGSEL) = selector;
    *rlb_reg(WIN_IOAPIC, IOAPIC_IOWIN)     = value;
}

/* Redirection entries are two 32-bit halves per pin. */
static unsigned rte_lo(unsigned irq) { return 0x10 + irq * 2; }
static unsigned rte_hi(unsigned irq) { return 0x11 + irq * 2; }

void ioapic_arm(unsigned irq, uint8_t vector, int masked)
{
    ioapic_write(rte_lo(irq), vector | ((masked ? 1u : 0u) << 16));
    ioapic_write(rte_hi(irq), 0); /* destination */
}
```

**Redirection entries reset masked.** This is the step most often missed, and its failure mode is the worst kind: the source reaches the IOAPIC's pin, the pin is correct, and no message is ever delivered. That is indistinguishable from a broken route unless you know to check the mask bit.

On vectors: the assignment is yours. The published typical-assignment table suggests 0x20 through 0x2F for the legacy lines. The integration test suite uses `0x40 + irq`, which is a **test convention, not a hardware property** - convenient because the vector then says which line produced it. Nothing in the RTL constrains the choice.

## 9.9 Step 5: Enable the Blocks

---

Per-block, from that block's specification. The only integration-level rule is that this comes last.

If you enable the HPET's legacy-replacement mode, be aware it takes over IRQ0 and IRQ8: HPET timers 0 and 1 then drive `hpet_legacy_irq0` and `hpet_legacy_irq8` into those lines and are suppressed on `hpet_timer_irq`. You should silence the 8254 channel 0 and the RTC periodic interrupt to match, or two sources will be competing for one line - legally, since the fabric ORs them, but confusingly.

## 9.10 Step 6: Verify

| Check                      | How                                                                                                                 |
|----------------------------|---------------------------------------------------------------------------------------------------------------------|
| Decode is sound            | All ten probes read without error; an unmapped address sets PSLVERR                                                 |
| Controllers initialised    | PIC_STATUS bit 0 set on each configured controller                                                                  |
| Cascade consistent         | Master IR2 follows the slave's INT. It should rise for an IRQ8-15 source - that is the cascade working, not a fault |
| A route works              | Assert a source, confirm pic_int_out, and confirm an IOAPIC delivery with the vector you programmed                 |
| Anything at all is pending | rlb_irq_out is asserted while any block interrupt is                                                                |

## 9.11 Clearing an Interrupt

In edge-triggered mode `pic_int_out` latches until the controller is acknowledged. Software should acknowledge and issue EOI normally.

The integration test suite instead resets the whole subsystem between phases, because the acknowledge and EOI paths carry their own block-level limitations that would make an integration test fail on somebody else's known defect. That is a verification convenience, not guidance for firmware. It is mentioned here only because it explains a real consequence: **a reset drops all controller and IOAPIC configuration**, so anything that resets must redo steps 3 and 4.

## 9.12 Related Documents

- [Use Cases](#) - complete routing paths, end to end
- [Architecture](#) - what the fabric does with a source
- [Window Map](#) - where each block's registers are specified
- [pic\\_8259\\_mas](#), [ioapic\\_mas](#) - the controllers' own specifications

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## 10 RLB Top - Use Cases

Six worked paths through the subsystem. Each one is exercised by the integration test suite, so each is known to work as described.

### 10.1 Use Case 1: A UART Interrupt Through the Master Controller

The simplest routing path: a source on a master line.

| Stage                                                                                                                                                                                                                                             | Signal                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| Source                                                                                                                                                                                                                                            | uart_irq from u_uart              |
| Fabric line                                                                                                                                                                                                                                       | IRQ4 (IRQ_UART, convention: COM1) |
| Master input                                                                                                                                                                                                                                      | w_master_pic_irq[4]               |
| Output                                                                                                                                                                                                                                            | pic_int_out                       |
| <pre>pic_init_cascade(0x20, 0x28);          /* or pic_init_single(0x20) -- IRQ4 is a master line */ uart_enable_interrupts();              /* see uart_16550_mas */ /* uart_irq -&gt; w_fabric_irq[4] -&gt; master IR4 -&gt; pic_int_out */</pre> |                                   |

The vector the CPU sees is master\_base + 4, so 0x24 with a base of 0x20.

### 10.2 Use Case 2: A GPIO Interrupt Through the Cascade

A source on a slave line. This is the common case, not the exception - four of the six sourcing blocks land on the slave controller.

| Stage                                                                                                                                               | Signal                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Source                                                                                                                                              | gpio_irq from u_gpio                              |
| Fabric line                                                                                                                                         | IRQ11 (IRQ_GPIO, a <b>choice</b> , parameterised) |
| Slave input                                                                                                                                         | slave IR3, because $11 - 8 = 3$                   |
| Slave output                                                                                                                                        | w_spic_int                                        |
| Master input                                                                                                                                        | w_master_pic_irq[2], forced from the slave        |
| Output                                                                                                                                              | pic_int_out, returning the <b>slave's</b> vector  |
| <pre>pic_init_cascade(0x20, 0x28);          /* mandatory: IRQ11 is a slave line */ gpio_enable_interrupt(pin);             /* see gpio_mas */</pre> |                                                   |

The vector is `slave_base + 3`, so `0x2B` with a slave base of `0x28`. The master returns the slave's vector in place of its own - that is what the cascade acknowledge path is for.

Master IR2 rising here is **correct behaviour**, not a spurious cascade interrupt. The invariant to check is that IR2 equals the slave's INT, never that IR2 stays low.

### 10.3 Use Case 3: The Same Source Through the IOAPIC

---

The fabric feeds the IOAPIC in parallel with the controllers. No source change is needed - only IOAPIC configuration.

```
ioapic_arm(11, 0x40 + 11, /* masked = */ 0); /* unmask IRQ11's entry */
gpio_enable_interrupt(pin);
/* Delivery appears on ioapic_irq_out_valid with vector 0x4B. */
```

Both paths are live at once: the same GPIO assertion reaches the cascade *and* produces an IOAPIC delivery. If you want only one, mask the other - mask the IOAPIC entry, or mask the line in the controller's OCW1.

Remember the mask default. An entry you never programmed is masked, and a masked entry delivers nothing while looking perfectly correct from the source side.

### 10.4 Use Case 4: HPET Legacy Replacement

---

Replacing the 8254 tick and the RTC periodic interrupt with HPET timers.

| HPET output                                                                                                                                                                   | Line | Replaces               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------------------|
| hpet_legacy_irq0                                                                                                                                                              | IRQ0 | 8254 channel 0 tick    |
| hpet_legacy_irq8                                                                                                                                                              | IRQ8 | RTC periodic interrupt |
| <pre>hpet_set_legacy_replacement(1); /* see hpet_mas */ pit_disable_counter(0); /* avoid two sources on IRQ0 */ rtc_disable_periodic(); /* avoid two sources on IRQ8 */</pre> |      |                        |

The HPET core suppresses timers 0 and 1 on `hpet_timer_irq` while this mode is active, so a single event cannot arrive twice from the HPET itself. Silencing the 8254 and the RTC is still your job: the fabric ORs all three onto those lines, so leaving them enabled is legal and produces interrupts from sources you thought you had replaced.

The general HPET timers are unaffected, and remain unrouted - they reach `rlb_irq_out` and their own pins only.

## 10.5 Use Case 5: Boot-Interrupt Rerouting

---

Catching a source whose IOAPIC entry is masked, so it is not lost during boot before the IOAPIC is configured.

```
ioapic_arm(irq, vector, /* masked = */ 1);    /* IOAPIC will NOT
deliver */
ioapic_write(IOAPIC_BOOTINTX, 1);             /* reroute masked pins
to the PIC */
/* The source now reaches the legacy controller instead. */
```

The map is the identity mapping for IOAPIC pins 0-7 onto legacy inputs 0-7, and no reroute above pin 7.

**Pin 2 cannot be rerouted.** The map still names legacy input 2, but master IR2 is masked off every external source and driven from the slave controller, so a reroute onto pin 2 reaches nothing. If you need pin 2's source on the legacy path, pick a different legacy input.

Rerouting is disabled at reset, so the master's interrupt input is exactly `pic_irq_in` until you perform the second write above.

## 10.6 Use Case 6: One Interrupt Line for the Whole Subsystem

---

For a system that would rather not model ten interrupt sources.

```
/* Wire rlb_irq_out to your controller. On assertion, poll to find the
source. */
if (rlb_irq_out_asserted()) {
    /* Check each block's status register to identify the source. */
}
```

`rlb_irq_out` asserts while **any** block interrupt is asserted, including `pic_int_out` itself. It carries no vector and no priority - it is a pure OR, and identifying the source means polling. It drives nothing inside the subsystem, which is precisely why including the controller's own output in it is safe.

This is also the path that sees the general HPET timers, which no other path does.

## 10.7 Coincident Sources

---

The fabric is combinational and per-line, so simultaneous sources are handled by construction: each lands on its own line, and lines that share a source are OR-ed. The integration suite exercises two, three and four coincident sources, spanning both controllers - for example the UART on master IR4 while SMBus, ACPI and GPIO assert together under the cascade.

Two practical notes from that testing:

- **Some sources are slow to assert.** ACPI takes tens of clocks from pm\_gpe\_events to pm\_interrupt, and SMBus considerably longer. Code that expects a source to be asserted a few cycles after its stimulus will sample too early.
- **Priority is the controller's job, not the fabric's.** The fabric presents every asserted line at once; which one the CPU services is decided by the 8259 priority logic, or by the IOAPIC's delivery configuration.

## 10.8 Related Documents

- [Initialization](#) - the bring-up sequences these use cases assume
- [Architecture](#) - the fabric, cascade and aggregate
- [Window Map](#)

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# 11 RLB Top - Address Map

## 11.1 rlb\_top Has No Registers

rlb\_top implements no registers of its own. Every address it answers belongs to a block it instantiates. This chapter is therefore the **window map** - how an address is decoded, and which specification documents the registers inside each window.

## 11.2 Decode

| Property              | Value                               |
|-----------------------|-------------------------------------|
| Base                  | BASE_ADDR, default 0xFEC0_0000      |
| Window size           | 0x1000 (4KB)                        |
| Windows               | 10, indices 0-9                     |
| Slave index           | PADDR[15:12]                        |
| Offset within a block | PADDR[11:0]                         |
| Reserved              | none - all ten windows are assigned |

| Property | Value                                 |
|----------|---------------------------------------|
| Unmapped | completes with PSLVERR; does not hang |

The decode is a clean power of two: the window number is  $PADDR[15:12]$ , and the address a block sees is the low twelve bits. A block's own register map is therefore documented in offsets from 0x000, and the subsystem address is  $BASE\_ADDR + window * 0x1000 + offset$ .

## 11.3 Window Map

| Window | Address range             | Block             | Register map                                                                      |
|--------|---------------------------|-------------------|-----------------------------------------------------------------------------------|
| 0      | 0xFEC0_0000 - 0xFEC0_0FFF | HPET              | <a href="#">hpet_mas</a><br><a href="#">ch05_registers/01_register_map.md</a>     |
| 1      | 0xFEC0_1000 - 0xFEC0_1FFF | 8259 PIC (master) | <a href="#">pic_8259_mas</a><br><a href="#">ch05_registers/01_register_map.md</a> |
| 2      | 0xFEC0_2000 - 0xFEC0_2FFF | 8254 PIT          | <a href="#">pit_8254_mas</a><br><a href="#">ch05_registers/01_register_map.md</a> |
| 3      | 0xFEC0_3000 - 0xFEC0_3FFF | RTC               | <a href="#">rtc_mas</a><br><a href="#">ch05_registers/01_register_map.md</a>      |
| 4      | 0xFEC0_4000 - 0xFEC0_4FFF | SMBus             | <a href="#">smbus_mas</a><br><a href="#">ch05_registers/01_register_map.md</a>    |
| 5      | 0xFEC0_5000 - 0xFEC0_5FFF | PM/ACPI           | <a href="#">pm_acpi_mas</a><br><a href="#">ch05_registers/01_register_map.md</a>  |
| 6      | 0xFEC0_6000 - 0xFEC0_6FFF | IOAPIC            | <a href="#">ioapic_mas</a><br><a href="#">ch05_registers/01_register_map.md</a>   |
| 7      | 0xFEC0_7000 - 0xFEC0_7FFF | GPIO              | <a href="#">gpio_mas</a><br><a href="#">ch05_registers/01_register_map.md</a>     |
| 8      | 0xFEC0_8000 - 0xFEC0_8FFF | UART 16550        | <a href="#">uart_16550_mas</a><br><a href="#">ch05_registers/01_</a>              |

| Window | Address range             | Block                    | Register map                                                                                    |
|--------|---------------------------|--------------------------|-------------------------------------------------------------------------------------------------|
| 9      | 0xFEC0_9000 - 0xFEC0_9FFF | 8259 PIC (cascade slave) | <a href="#">register_map.md</a><br><a href="#">pic_8259_mach05_registers/01_register_map.md</a> |

Windows 1 and 9 are the same block with the same register map, wired as a master and slave pair. The registers are identical; what differs is the cascade configuration written into them. See [Initialization](#).

## 11.4 Read-Safe Probe Registers

For confirming a window is alive without disturbing it.

| Window | Block        | Probe offset | Register                                  |
|--------|--------------|--------------|-------------------------------------------|
| 0      | HPET         | 0x000        | HPET_ID, read-only identity               |
| 1      | PIC (master) | 0x000        | PIC_CONFIG                                |
| 2      | PIT          | 0x000        | PIT_CONFIG                                |
| 3      | RTC          | 0x000        | RTC_CONFIG                                |
| 4      | SMBus        | 0x000        | SMBUS_CONTROL                             |
| 5      | PM/ACPI      | 0x000        | ACPI_CONTROL                              |
| 6      | IOAPIC       | 0x000        | IOREGSEL                                  |
| 7      | GPIO         | 0x000        | GPIO_CONTROL                              |
| 8      | UART 16550   | <b>0x020</b> | UART_SCR<br>scratch - <b>not</b><br>0x000 |
| 9      | PIC (slave)  | 0x000        | PIC_CONFIG                                |

**The UART is the exception and it matters.** Offset 0x000 in the UART window is the receive buffer; reading it pops a byte off the RX FIFO. Probe the scratch register at 0x020 instead. An identity read that quietly consumes received data is a genuinely difficult bug to find.

## 11.5 The IOAPIC Is Indirect

The IOAPIC occupies a 4KB window but exposes only two directly addressed registers. Everything else is reached by writing an index to IOREGSEL and then the data to IOWIN. The redirection table in particular is two 32-bit halves per pin, reached through that indirection - so an IOAPIC “register offset” in the IOAPIC specification is a selector value, not a window offset.

## 11.6 Interrupt Line Assignments

---

Not registers, but the other half of the programming surface. The line numbers are fixed in hardware for convention assignments and parameterised for choices.

| Line  | Source                                                       | Basis                                  | Controller input                    |
|-------|--------------------------------------------------------------|----------------------------------------|-------------------------------------|
| IRQ0  | pit_timer_irq[0]<br>OR<br>hpet_legacy_irq0                   | convention                             | master IR0                          |
| IRQ2  | cascade only -<br>never driven by<br>the fabric              | convention                             | master IR2, from<br>the slave's INT |
| IRQ4  | uart_irq                                                     | convention                             | master IR4                          |
| IRQ8  | rtc_alarm_irq OR<br>rtc_second_irq<br>OR<br>hpet_legacy_irq8 | convention                             | slave IR0                           |
| IRQ9  | pm_interrupt                                                 | convention                             | slave IR1                           |
| IRQ10 | smb_interrupt                                                | <b>choice</b> , parameter<br>IRQ_SMBUS | slave IR2                           |
| IRQ11 | gpio_irq                                                     | <b>choice</b> , parameter<br>IRQ_GPIO  | slave IR3                           |

Each line also reaches the IOAPIC pin of the same number. The IOAPIC's vector for a pin is whatever software programs into that pin's redirection entry; there is no hardware-fixed vector.

## 11.7 Related Documents

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- [Top-Level Interface](#) - the decode contract and port list
- [Initialization](#) - the address helper and bring-up order
- [Architecture](#) - how a line reaches its destinations

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## 12 Retro Legacy Blocks - Product Requirements Document

**Component:** Retro Legacy Blocks (RLB) - Production-Quality Legacy Peripherals **Version:** 1.0  
**Status:** Active Development - HPET Production Ready **Last Updated:** 2025-10-29

---

### 12.1 1. Overview

---

#### 12.1.1 1.1 Purpose

The Retro Legacy Blocks (RLB) component provides production-quality implementations of legacy peripheral blocks based on proven peripheral designs. These blocks are designed to be reusable, well-tested, and suitable for both FPGA and ASIC implementation.

#### 12.1.2 1.2 Design Philosophy

**“Retro” - Proven Architectures:** - Implements time-tested peripheral designs from successful platforms - Focuses on simplicity, reliability, and well-understood behavior - Prioritizes production-readiness over experimental features

**“Legacy” - Time-Tested Interfaces:** - Based on proven peripheral interface specifications - Suitable for systems requiring retro-compatible peripheral compatibility - APB-based interface for easy integration

**“Blocks” - Modular Collection:** - Each peripheral is independent and self-contained - Clear separation between different blocks (rtl/hpet/, rtl/gpio/, etc.) - Can be used individually or wrapped into integrated subsystem

#### 12.1.3 1.3 Target Applications

- Retro-compatible platform compatibility layers
  - Embedded systems requiring legacy peripheral interfaces
  - FPGA-based system emulation
  - Educational platforms demonstrating classic peripheral designs
  - Mixed-vintage SoC integration (modern + legacy interfaces)
-

## 12.2 2. Implemented Blocks

---

### 12.2.1 12.1 HPET - High Precision Event Timer

**Status:** Production Ready (5/6 configurations 100% passing) **RTL Location:** rtl/hpet/

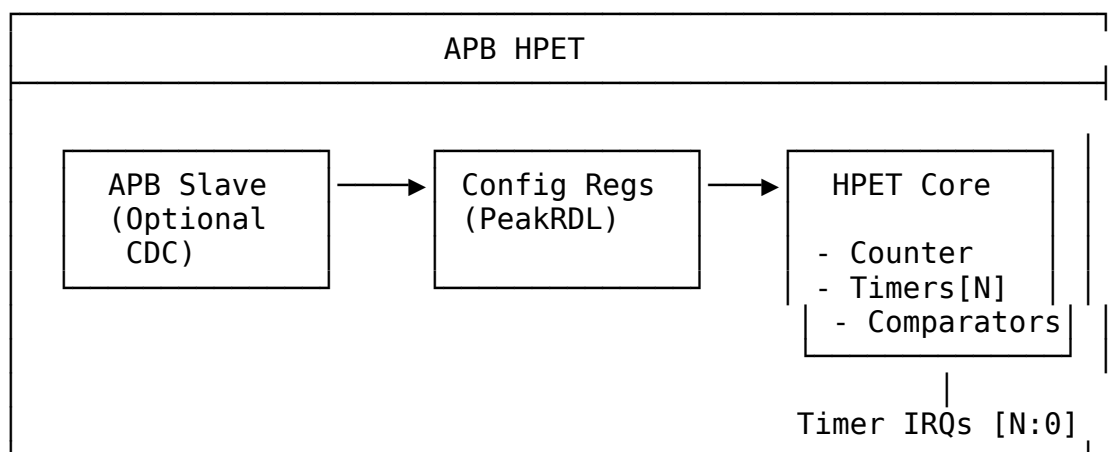
**Documentation:** docs/hpet\_mas/

**Key Features:** - Configurable timer count: 2, 3, or 8 independent timers - 64-bit main counter for high-resolution timestamps - 64-bit comparators per timer - Operating modes: One-shot and periodic - Clock domain crossing: Optional CDC for timer/APB clock independence - APB4 interface: Standard AMBA APB protocol - PeakRDL integration: Register map generated from SystemRDL specification

**Applications:** - System tick generation - Real-time OS scheduling - Precise event timing - Performance profiling - Watchdog timers - Multi-rate timing domains

**Test Coverage:** - 6 configurations tested (2/3/8 timers, CDC on/off) - 5/6 configurations at 100% pass rate - 1 configuration at 92% (minor stress test timeout) - 12 test cases per configuration (gate/func/full)

**Architecture:**



**Design Highlights:** - Reset macro standardization (FPGA-friendly) - Per-timer data buses prevent corruption - Edge-triggered register write strobes (not level) - W1C status register for interrupt clearing - Optional asynchronous clock domains with handshake CDC

**See:** docs/hpet\_mas/hpet\_mas\_index.md for complete HPET specification

---

## 12.3 3. Planned Blocks

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### 12.3.1 3.1 8259 - Programmable Interrupt Controller (PIC)

**Status:** Implemented (see dv/tests and the block MAS; per-block defects tracked in known\_issues/)

**Priority:** High **Address:** 0xFEC0\_1000 - 0xFEC0\_1FFF (4KB window, master of the cascaded pair)

**Planned Features:** - 8259A-compatible register interface - 8 interrupt request (IRQ) inputs - Cascadable (master/slave configuration) - Priority resolver (fixed and rotating priority) - Edge and level triggered modes - Interrupt mask register - End-of-Interrupt (EOI) handling - APB register interface

**Applications:** - Legacy interrupt management - PC-compatible systems - Hardware interrupt aggregation - Priority-based interrupt handling - Cascaded multi-level interrupt systems

### 12.3.2 3.2 8254 - Programmable Interval Timer (PIT)

**Status:** Implemented (see dv/tests and the block MAS; per-block defects tracked in known\_issues/)

**Priority:** High **Address:** 0xFEC0\_2000 - 0xFEC0\_2FFF (4KB window)

**Planned Features:** - 8254-compatible register interface - 3 independent 16-bit counters - 6 programmable counter modes - Binary and BCD counting - Read-back command - Configurable clock input - Interrupt/output generation per counter - APB register interface

**Counter Modes:** - Mode 0: Interrupt on terminal count - Mode 1: Hardware retriggerable one-shot - Mode 2: Rate generator - Mode 3: Square wave mode - Mode 4: Software triggered strobe - Mode 5: Hardware triggered strobe

**Applications:** - System tick generation - Periodic timer interrupts - Square wave generation - Event counting - Legacy PC timer compatibility

### 12.3.3 3.3 GPIO - General Purpose I/O

**Status:** Implemented (see dv/tests and the block MAS; per-block defects tracked in known\_issues/)

**Priority:** Medium **Address:** 0xFEC0\_7000 - 0xFEC0\_7FFF (4KB window)

**Planned Features:** - Configurable pin count (8, 16, 32 pins) - Per-pin direction control (input/output/bidirectional) - Input debouncing logic - Interrupt generation (rising/falling/both edges, level) - Output drive strength configuration - Pull-up/pull-down control - APB register interface

**Applications:** - LED control - Button inputs - Hardware control signals - Chip-select generation - Status monitoring

### 12.3.43.4 RTC - Real-Time Clock

**Status:** Implemented (apb4\_rtc + MAS; see docs/rtc\_mas – binary 24-hour timekeeping validated; BCD/12-hour modes and the production-mode time-set path are broken, tracked in issue #56)

**Priority:** Medium **Effort:** 3-4 weeks **Address:** 0xFEC0\_3000 - 0xFEC0\_3FFF (4KB window)

**Planned Features:** - 32.768 kHz clock input (typical RTC crystal frequency) - Seconds, minutes, hours, day, month, year tracking - Alarm functionality - Battery backup support (power domain considerations) - 24-hour or 12-hour (AM/PM) mode - Leap year handling - APB register interface

**Applications:** - System time-of-day tracking - Wake-on-alarm functionality - Timestamp generation - Power-aware applications

### 12.3.53.5 SMBus Controller

**Status:** Implemented (see dv/tests and the block MAS; per-block defects tracked in known\_issues/)

**Priority:** Medium **Address:** 0xFEC0\_4000 - 0xFEC0\_4FFF (4KB window)

**Planned Features:** - SMBus 2.0 compliance - Master and slave modes - Clock stretching support - Packet Error Checking (PEC) - Alert response address - Configurable clock speed - APB register interface

**Applications:** - System management bus communication - Sensor interfaces (temperature, voltage) - EEPROM access - Battery management - Fan control

### 12.3.63.6 UART - Universal Asynchronous Receiver/Transmitter

**Status:** Implemented (apb4\_uart\_16550 + MAS; see docs/uart\_16550\_mas – note the shipped block has 1/2 stop bits only and no automatic flow control, and issue #60 tracks its RTL defects) **Priority:** Medium **Address:** 0xFEC0\_8000 - 0xFEC0\_8FFF (4KB window)

**Planned Features:** - 16550-compatible register interface - Configurable baud rate generation - 5/6/7/8 data bits - Parity: none, even, odd, mark, space - Stop bits: 1, 1.5, 2 - Hardware flow control (RTS/CTS) - FIFO buffers (16-byte TX/RX) - Interrupt generation

**Applications:** - Debug console - Serial communication - Modem interfaces - Legacy peripheral communication

### 12.3.73.7 SPI Controller

**Status:** Planned **Priority:** Low **Effort:** 5-6 weeks **Address:** TBD (not in primary ILB address map)

**Planned Features:** - Master mode (initially; slave mode future) - Configurable clock polarity and phase (CPOL/CPHA) - Multiple chip selects - Configurable word size (8/16/32 bits) - TX/RX FIFOs - DMA support (future) - APB register interface

**Applications:** - Flash memory access - ADC/DAC interfaces - Display controllers - SD card communication

### 12.3.83.8 I2C Controller

**Status:** Planned **Priority:** Low **Effort:** 5-7 weeks **Address:** TBD (not in primary ILB address map)

**Planned Features:** - I2C standard (100 kHz), fast (400 kHz), fast-plus (1 MHz) modes - Multi-master arbitration - 7-bit and 10-bit addressing - Clock stretching - General call support - APB register interface

**Applications:** - Sensor interfaces - EEPROM access - Multi-chip communication - System configuration

### 12.3.93.9 Watchdog Timer

**Status:** Planned **Priority:** Low **Effort:** 2-3 weeks **Address:** TBD (not in primary ILB address map)

**Planned Features:** - Configurable timeout period - Countdown counter with reload - Reset generation on timeout - Lock mechanism to prevent accidental disable - Interrupt before reset (optional warning) - APB register interface

**Applications:** - System fault recovery - Software hang detection - Periodic system reset - Safety-critical applications

## 12.3.10 3.10 Power Management / ACPI Controller

**Status:** Implemented (see dv/tests and the block MAS; per-block defects tracked in known\_issues/)

**Priority:** Medium **Address:** 0xFEC0\_5000 - 0xFEC0\_5FFF (4KB window)

**Planned Features:** - Clock gating control per block - Power domain sequencing - Reset generation and distribution - Wake event handling - Sleep/idle mode control - ACPI-compatible registers - APB register interface

**Applications:** - Low-power system design - Battery-powered devices - Dynamic power management - Thermal management - OS power management interface

## 12.3.11 3.11 IOAPIC - I/O Advanced Programmable Interrupt Controller

**Status:** Implemented (see dv/tests and the block MAS; per-block defects tracked in known\_issues/)

**Priority:** Medium **Address:** 0xFEC0\_6000 - 0xFEC0\_6FFF (4KB window)

**Planned Features:** - I/O APIC CSR model (register-based interface) - Multiple interrupt inputs (24+) - Programmable interrupt routing - Edge and level triggered modes - Priority-based arbitration - Interrupt masking per input - APB register interface for configuration

**Applications:** - Advanced interrupt routing - Multi-processor interrupt distribution - Flexible interrupt mapping - Legacy IRQ redirection - PC-compatible systems

## 12.3.12 3.12 Interconnect ID / Version Registers

**Status:** Planned – NOT present in the shipped design **Priority:** Low **Effort:** 1-2 weeks **Address:** unallocated. The shipped decoder is the generated apbx\_xbar\_1to10, whose ten windows

(0x0000-0x9FFF) are fully assigned, so adding this block needs a window assignment AND a crossbar regeneration – it cannot simply occupy 0xF000 as earlier drafts of this document assumed.

**Planned Features:** - Vendor ID register - Device ID register - Revision ID register - Block presence/capability bits - Configuration status registers - Debug/diagnostic registers - APB register interface

**Applications:** - Software block discovery - Version checking - Feature detection - Debug and diagnostics - Platform identification

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## 12.4 4. Integration and Wrapper Goals

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### 12.4.1 4.1 Individual Block Integration

Each block is designed to be used standalone:

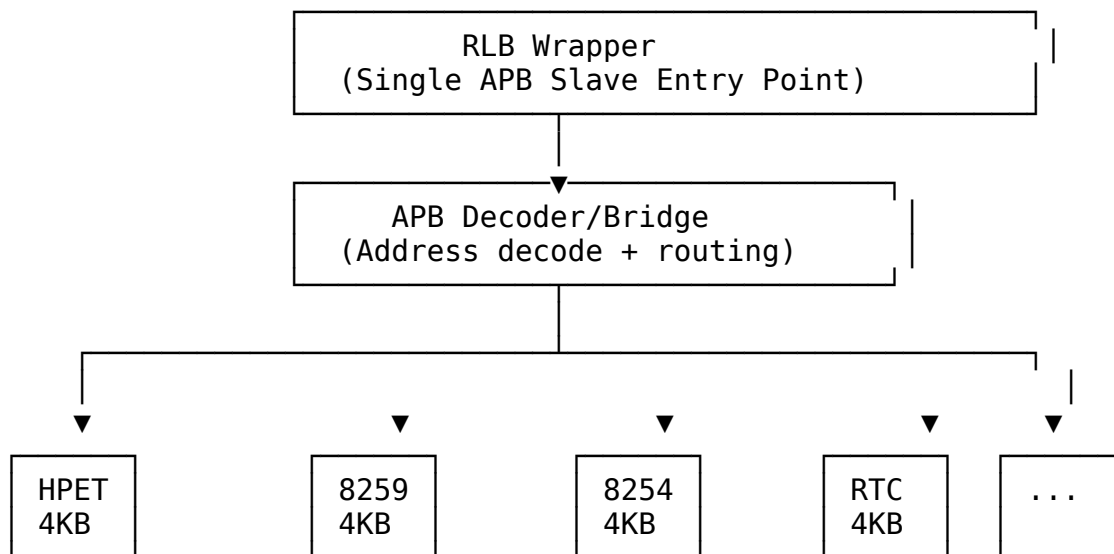
**Example - HPET Integration:**

```
apb4_hpet #(
    .NUM_TIMERS(3),
    .VENDOR_ID(16'h4747),
    .REVISION_ID(16'h0001),
    .CDC_ENABLE(0)
) u_hpet (
    .pclk            (apb_clk),
    .presetn         (apb_rst_n),
    // APB interface -- the ports are s_apb_P*, not bare p*
    .s_apb_PSEL      (psel_hpet),
    .s_apb_PENABLE   (penable),
    .s_apb_PREADY    (pready_hpet),
    .s_apb_PADDR     (paddr[11:0]),
    .s_apb_PWRITE    (pwrite),
    .s_apb_PWDATA    (pwwdata),
    .s_apb_PSTRB     (pstrb),
    .s_apb_PPROT     (pprot),
    .s_apb_PRDATA    (prdata_hpet),
    .s_apb_PSLVERR   (pslverr_hpet),
    // HPET-specific
    .hpet_clk        (timer_clk),
    .hpet_resetsn    (timer_rst_n),
    .timer_irq       (timer_irq[2:0])
);
```

### 12.4.24.2 RLB Wrapper Architecture

**Goal:** Create top-level wrapper combining multiple legacy blocks into unified retro-compatible subsystem.

**System Architecture:**



**Address Map:**

Base address: 0xFEC0\_0000 – the BASE\_ADDR parameter default in `rlb_top.sv`, and the conventional location for this class of block. It is a PARAMETER: an integrator may relocate the whole window set. Window size: 4KB per block (clean power-of-2 decode)

| Address Range             | Block             | Size | Function                          |
|---------------------------|-------------------|------|-----------------------------------|
| 0xFEC0_0000 - 0xFEC0_0FFF | HPET              | 4KB  | High Precision Event Timer        |
| 0xFEC0_1000 - 0xFEC0_1FFF | 8259 PIC (master) | 4KB  | Programmable Interrupt Controller |
| 0xFEC0_2000 - 0xFEC0_2FFF | 8254 PIT          | 4KB  | Programmable Interval Timer       |
| 0xFEC0_3000 - 0xFEC0_3FFF | RTC               | 4KB  | Real-Time Clock                   |
| 0xFEC0_4000 - 0xFEC0_4FFF | SMBus             | 4KB  | SMBus Host Controller             |
| 0xFEC0_5000 - 0xFEC0_5FFF | PM/ACPI           | 4KB  | Power Management / ACPI           |

| Address Range             | Block             | Size | Function                                   |
|---------------------------|-------------------|------|--------------------------------------------|
|                           |                   |      | Registers                                  |
| 0xFEC0_6000 - 0xFEC0_6FFF | IOAPIC            | 4KB  | I/O Advanced PIC (CSR model)               |
| 0xFEC0_7000 - 0xFEC0_7FFF | GPIO              | 4KB  | General Purpose I/O                        |
| 0xFEC0_8000 - 0xFEC0_8FFF | UART 16550        | 4KB  | Serial port                                |
| 0xFEC0_9000 - 0xFEC0_9FFF | 8259 PIC (slave)  | 4KB  | Cascade slave of the PC/AT pair            |
| All other addresses       | decode-miss agent | -    | Accepted and answered locally with PSLVERR |

The decoder is the GENERATED apbx\_xbar\_1to10 (one master, ten slaves), so the window count and the base both come from the generator invocation recorded in rlb\_top.sv. There is no window above 0x9FFF.

#### Decoder Implementation:

```
// Address decode logic (simplified)
localparam BASE_ADDR = 32'hFEC0_0000;
localparam BLOCK_SIZE = 12; // 4KB = 2^12

logic [3:0] block_sel;
assign block_sel = paddr[15:12]; // Extract window number

always_comb begin
    psel_hpet      = (block_sel == 4'h0) & psel; // 0xFEC0_0xxx
    psel_pic8259   = (block_sel == 4'h1) & psel; // 0xFEC0_1xxx
    psel_pit8254   = (block_sel == 4'h2) & psel; // 0xFEC0_2xxx
    psel_rtc       = (block_sel == 4'h3) & psel; // 0xFEC0_3xxx
    psel_smbus     = (block_sel == 4'h4) & psel; // 0xFEC0_4xxx
    psel_pm        = (block_sel == 4'h5) & psel; // 0xFEC0_5xxx
    psel_ioapic    = (block_sel == 4'h6) & psel; // 0xFEC0_6xxx
    psel_gpio      = (block_sel == 4'h7) & psel; // 0xFEC0_7xxx
    psel_uart      = (block_sel == 4'h8) & psel; // 0xFEC0_8xxx
    psel_pic_slave = (block_sel == 4'h9) & psel; // 0xFEC0_9xxx
    psel_error     = !((psel_hpet, psel_pic8259, psel_pit8254,
                       psel_rtc, psel_smbus, psel_pm, psel_ioapic,
                       psel_gpio, psel_uart, psel_pic_slave) &
    psel;
end
```

**Interface:** - **Single APB slave port** at base address 0xFEC0\_0000 (the BASE\_ADDR parameter) - **Aggregated interrupt output** combining all block IRQs - **Per-block clock/reset control** for power management - **External I/O signals** (GPIO, UART, I2C/SMBus, etc.) - **Error slave** returns SLVERR for unmapped addresses

**Benefits:** - Simplified system integration (single APB slave) - Consistent 4KB window addressing - Clean power-of-2 address decode - Ten 4KB windows, fully assigned; expansion needs a crossbar regeneration - Single verification target - Drop-in retro-compatible peripheral subsystem

---

## 12.5 5. Design Standards

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### 12.5.15.1 Reset Handling

**MANDATORY:** All blocks must use standardized reset macros from rtl/amba/includes/reset\_defs.svh

**Pattern:**

```
`include "reset_defs.svh"

`ALWAYS_FF_RST(clk, rst_n,
  if (`RST_ASSERTED(rst_n)) begin
    r_state <= IDLE;
    r_counter <= '0;
  end else begin
    r_state <= w_next_state;
    r_counter <= r_counter + 1'b1;
  end
)
```

**Why:** - FPGA-friendly reset inference - Consistent synthesis behavior - Single-point reset polarity control - Better timing closure

### 12.5.25.2 Register Generation

**Preferred:** Use PeakRDL for register map generation

**Process:** 1. Define registers in SystemRDL (.rdl file) 2. Generate RTL using PeakRDL regblock 3. Create wrapper module connecting registers to core logic 4. Use edge detection for write strobes (not level)

**Benefits:** - Consistent register interface - Auto-generated documentation - Reduced manual RTL errors - Easy register map changes

### 12.5.35.3 Testbench Architecture

**MANDATORY:** Follow project testbench organization pattern

### Structure:

```
dv/
├── tbclasses/{block}/          # Block-specific TB classes
│   ├── {block}_tb.py          # Main testbench
│   ├── {block}_tests_basic.py # Gate test suite
│   ├── {block}_tests_medium.py # Func test suite
│   └── {block}_tests_full.py  # Full test suite
├── tests/{block}/             # Test runners
│   ├── test_apb4_{block}.py   # Pytest wrapper
│   └── conftest.py            # Pytest configuration
```

### Import Pattern:

```
# Always import from PROJECT AREA
from projects.components.retro_legacy_blocks.dv.tbclasses.{block}.
{block}_tb import {Block}TB
```

**Test Levels:** - **Gate:** Core functionality (register access, basic operation) - **Func:** Extended features (modes, configurations, edge cases) - **Full:** Stress testing, CDC variants, corner cases

**Target:** 100% pass rate at all levels

## 12.5.45.4 FPGA Synthesis Attributes

**MANDATORY:** Add FPGA synthesis hints for memory arrays

```
`ifdef XILINX
    (* ram_style = "auto" *)
`elsif SYNTH_PRAGMA
    /* synthesis ramstyle = "AUTO" */
`endif
logic [DATA_WIDTH-1:0] mem [DEPTH];
```

## 12.5.55.5 Documentation Requirements

Each block must have: - RTL comments (inline) - Register map specification - Block-level specification in docs/{block}\_mas/ - Integration guide - Test plan and results

---

## 12.6 6. Quality Metrics

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### 12.6.16.1 Production Readiness Criteria

A block is considered “Production Ready” when:

- All gate tests pass 100%
- All func tests pass 100%

- All full tests pass  $\geq 95\%$
- Complete register map specification
- RTL lint clean (Verilator)
- Reset macros used throughout
- FPGA synthesis attributes applied
- Integration guide written
- Known issues documented

### 12.6.26.2 Current Status

| Block    | Priority | Status      | Test Pass Rate       | Documentation | Production Ready                                          |
|----------|----------|-------------|----------------------|---------------|-----------------------------------------------------------|
| HPET     | High     | Implemented | see dv/tests         | MAS (ch1/2/5) | RTL defects open (#46)                                    |
| 8259 PIC | High     | Implemented | see dv/tests         | MAS (ch1/5)   | ISR/EOI half inert (#50)                                  |
| 8254 PIT | High     | Implemented | 6/6 both CDC configs | MAS (full)    | Mode 0 only (#52)                                         |
| GPIO     | Medium   | Implemented | see dv/tests         | MAS (full)    | Functional; quirks tracked (#44)                          |
| RTC      | Medium   | Implemented | see dv/tests         | MAS (ch1/5)   | Binary 24h only; time-set broken in production mode (#56) |
| SMBus    | Medium   | Implemented | see dv/tests         | MAS (ch1/5)   | Transfer engine non-functional (#58)                      |
| PM/ACPI  | Medium   | Implemented | see dv/tests         | MAS (ch1/5)   | Status/W1C/wake defects (#54)                             |
| IOAPIC   | Medium   | Implemented | see dv/tests         | MAS (ch1/2/5) | Double delivery + EOI hazards (#48)                       |

| Block        | Priority | Status      | Test Pass Rate | Documentation | Production Ready              |
|--------------|----------|-------------|----------------|---------------|-------------------------------|
| UART         | Medium   | Implemented | see dv/tests   | MAS (full)    | FE/BI dead, IER unwired (#60) |
| SPI          | Low      | Planned     | N/A            | N/A           | No                            |
| I2C          | Low      | Planned     | N/A            | N/A           | No                            |
| Watchdog     | Low      | Planned     | N/A            | N/A           | No                            |
| Interconnect | Low      | Planned     | N/A            | N/A           | No                            |

## 12.7 7. Development Roadmap

### 12.7.17.1 Phase 1: Foundation (Complete)

- HPET implementation
- Directory structure for multiple blocks
- Testbench architecture established
- Documentation templates
- Build and test infrastructure

### 12.7.27.2 Phase 2: Core Peripherals (Next 6-9 Months)

**Q1 2026 (High Priority):** - 8259 PIC (6-8 weeks) - Interrupt controller - 8254 PIT (4-5 weeks) - Interval timer - RTC (3-4 weeks) - Real-time clock

**Q2 2026 (Medium Priority):** - GPIO Controller (4-6 weeks) - SMBus Controller (6-8 weeks) - PM/ACPI Controller (8-10 weeks)

**Q3 2026:** - UART (4-5 weeks) - IOAPIC (6-8 weeks)

### 12.7.37.3 Phase 3: Advanced Peripherals (9-15 Months)

**Q4 2026:** - SPI Controller (5-6 weeks) - I2C Controller (5-7 weeks) - Watchdog Timer (2-3 weeks)

**Q1 2027:** - Interconnect ID/Version Registers (1-2 weeks) - ILB Wrapper integration starts

## 12.7.47.4 Phase 4: System Integration (15+ Months)

**Q2-Q4 2027:** - Complete ILB wrapper with all blocks - System-level integration examples - Performance characterization - FPGA reference designs - Application notes - Software driver examples

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## 12.8 8. References

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### 12.8.18.1 External Standards

**Peripheral Specifications:** - ACPI HPET Specification 1.0a - SMBus Specification Version 2.0 - 16550 UART Datasheet - I2C Specification (NXP) - SPI Protocol Specification

**Bus Protocols:** - AMBA APB Protocol Specification (ARM) - AMBA 3 APB Protocol v1.0

### 12.8.28.2 Internal Documentation

- /CLAUDE.md - Repository AI guide
- /PRD.md - Master repository requirements
- projects/components/retro\_legacy\_blocks/CLAUDE.md - Component AI guide
- projects/components/retro\_legacy\_blocks/README.md - Component overview
- vault/Tasks/RLB/ - task tracking (cross-block); vault/Tasks/RLB/hpet/ for HPET

### 12.8.38.3 Block-Specific Documentation

**HPET:** - docs/hpet\_mas/hpet\_mas\_index.md - HPET specification - docs/IMPLEMENTATION\_STATUS.md - HPET test results - known\_issues/ - HPET issue tracking

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## 12.9 9. Success Criteria

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### 12.9.19.1 Individual Block Success

Each block must: - Pass all gate/func tests at 100% - Pass full tests at  $\geq 95\%$  - Have complete register map specification - Include integration guide with examples - Be lint-clean (Verilator) - Use reset macros throughout - Include FPGA synthesis attributes

### 12.9.29.2 Collection Success

The `retro_legacy_blocks` component is successful when: - At least 6 blocks production-ready (HPET + 5 high/medium priority blocks) - All blocks follow consistent architecture (reset macros, PeakRDL, APB interface) - RLB wrapper integrates all blocks seamlessly with clean 4KB addressing - System-level integration example provided - Complete documentation for all blocks - FPGA reference design available - Address map covers all essential retro-compatible peripherals

### 12.9.39.3 Long-Term Vision

Ultimate goal: - Production-quality retro-compatible peripheral subsystem - Complete peripheral coverage for legacy platform requirements - Used in production FPGA designs - Educational resource for classic peripheral design - Foundation for mixed-vintage SoC designs

---

**Version:** 1.0 **Last Review:** 2025-10-29 **Next Review:** After each new block completion **Maintained By:** RTL Design Sherpa Project

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## 13 RLB (Retro Legacy Blocks) - FPGA Implementation Guide

**Date:** 2025-11-16 **Purpose:** Explain RLB blocks and FPGA implementation compatibility

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### 13.1 What Are RLB (Retro Legacy Blocks)?

---

#### 13.1.1 Overview

**Retro Legacy Blocks** are modern implementations of classic PC/embedded system peripherals, designed as reusable IP cores with APB interfaces. They provide functionality compatible with legacy systems but implemented in clean, modern SystemVerilog.

#### 13.1.2 Purpose

**Educational:** - Learn how classic PC peripherals work internally - Understand interrupt controllers, timers, power management - Study register-based peripheral design - Practice verification of complex protocols

**Practical:** - Build soft-core SoC systems on FPGAs - Create retro computer replicas (PC-compatible)  
- Implement embedded systems with familiar peripherals - Rapid prototyping of control systems

**Industrial:** - Reusable IP cores for custom SoCs - Known-good peripheral implementations - Well-documented, tested components - Modern coding standards with legacy compatibility

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## 13.2 Implemented RLB Modules

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### 13.2.1 Complete Modules

#### 13.2.1.1 1. HPET (*High Precision Event Timer*)

- **Purpose:** High-resolution timer (sub-microsecond precision)
- **Features:** Multiple timers, one-shot/periodic modes, 64-bit counters
- **PC Equivalent:** Modern replacement for PIT
- **Use Cases:** Precise timing, event scheduling, benchmarking

#### 13.2.1.2 2. PIT\_8254 (*Programmable Interval Timer*)

- **Purpose:** Classic PC timer (3 channels)
- **Features:** Mode 0-5 operation, binary/BCD counting
- **PC Equivalent:** 8254 PIT
- **Use Cases:** System tick, speaker control, legacy PC compatibility

#### 13.2.1.3 3. RTC (*Real-Time Clock*)

- **Purpose:** Calendar and alarm functions
- **Features:** Time/date tracking, alarm, periodic interrupts
- **PC Equivalent:** MC146818 / DS12887
- **Use Cases:** Wall-clock time, wake-on-alarm, timestamping

#### 13.2.1.4 4. PIC\_8259 (*Programmable Interrupt Controller*)

- **Purpose:** Legacy interrupt management
- **Features:** 8 IRQ inputs, priority, cascading, masking
- **PC Equivalent:** 8259A PIC
- **Use Cases:** Legacy PC interrupts, simple systems

#### 13.2.1.5 5. SMBus (*System Management Bus*)

- **Purpose:** Low-speed system management communication
- **Features:** I2C/SMBus 2.0, master/slave, PEC, FIFOs
- **PC Equivalent:** Modern motherboard SMBus controller

- **Use Cases:** Sensor monitoring, battery management, system config

#### 13.2.1.6 6. PM ACPI (Power Management ACPI)

- **Purpose:** Advanced power management
- **Features:** PM timer, power states (S0/S1/S3), GPE, clock gating, power domains
- **PC Equivalent:** ACPI PM controller
- **Use Cases:** Low-power systems, battery devices, sleep/wake, clock control

#### 13.2.1.7 7. IOAPIC (I/O Advanced Programmable Interrupt Controller)

- **Purpose:** Advanced interrupt routing
  - **Features:** 24 IRQs, redirection table, edge/level, priority, EOI
  - **PC Equivalent:** 82093AA I/O APIC
  - **Use Cases:** Multi-processor systems, flexible IRQ routing, modern PCs
- 

## 13.3 FPGA Implementation - Nexys A7 / Genesys2

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### 13.3.1 Can RLB Be Implemented on These Boards? YES!

All RLB modules are **100% compatible** with Nexys A7 and Genesys2 FPGAs.

### 13.3.2 Target FPGA Specifications

#### 13.3.2.1 Nexys A7 (Artix-7)

- **FPGA:** XC7A100T-1CSG324C (or XC7A50T)
- **Logic Cells:** 101,440 (A100T) or 52,160 (A50T)
- **Block RAM:** 4,860 Kb
- **Clock:** 100 MHz oscillator
- **I/O:** Extensive (PMOD, switches, LEDs, UART, etc.)
- **Verdict:** Excellent fit for full RLB system

#### 13.3.2.2 Genesys 2 (Kintex-7)

- **FPGA:** XC7K325T-2FFG900C
- **Logic Cells:** 326,080
- **Block RAM:** 16,020 Kb
- **Clock:** 200 MHz oscillator
- **I/O:** Very extensive
- **Verdict:** More than sufficient, ideal for complex SoC

### 13.3.3 Resource Usage Estimates

Per Module (Approximate):

| Module   | LUTs      | FFs      | BRAM | Notes                        |
|----------|-----------|----------|------|------------------------------|
| HPET     | ~800-1200 | ~600-900 | 0    | Multi-timer, 64-bit counters |
| PIT_8254 | ~400-600  | ~300-500 | 0    | 3 channels, modes            |
| RTC      | ~300-500  | ~250-400 | 0    | Time/date logic              |
| PIC_8259 | ~200-400  | ~150-300 | 0    | Simple priority logic        |
| SMBus    | ~600-900  | ~500-700 | 0.5  | FIFOs, protocol FSM          |
| PM_ACPI  | ~500-800  | ~400-600 | 0    | Power FSM, GPE               |
| IOAPIC   | ~800-1200 | ~600-900 | 0    | 24 IRQs, redirection         |

**Total RLB System (~5500 LUTs, ~4300 FFs):** - Nexys A7-100T: ~5% logic utilization Plenty of room  
- Nexys A7-50T: ~11% logic utilization Still comfortable - Genesys 2: ~2% logic utilization Massive headroom

### 13.3.4 What Can Fit on These Boards?

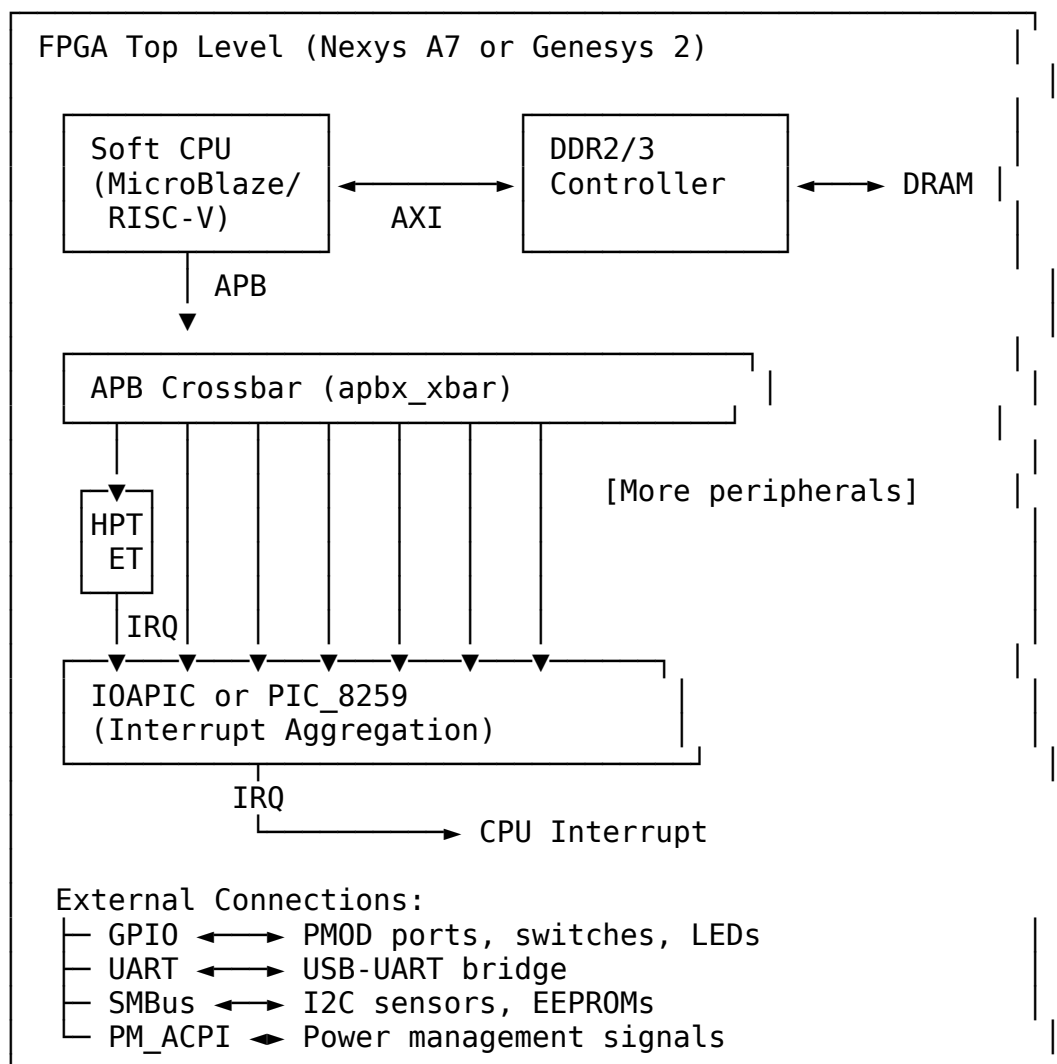
**Nexys A7-100T Can Support:** - All 7 RLB modules - MicroBlaze or RISC-V soft processor - DDR2 controller - Ethernet MAC - VGA/HDMI display controller - USB interface - Audio codec - **Result:** Complete retro-PC or modern embedded system

**Genesys 2 Can Support:** - Everything above PLUS: - Multiple soft processors - PCIe endpoint - High-speed interfaces - Complex video processing - **Result:** Professional-grade SoC development platform

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## 13.4 RLB System Integration on FPGA

### 13.4.1 Typical FPGA System Architecture



### 13.4.2 Example Use Cases on FPGAs

#### 13.4.2.1 1. Retro PC Replica

Build a PC-compatible system on FPGA: - Soft x86 core (or compatible RISC-V with x86 emulation) - All RLB peripherals for PC compatibility - VGA output for display - PS/2 keyboard/mouse - Run DOS, early Windows, Linux

### 13.4.2.2 2. Embedded Control System

Industrial/automotive control: - RISC-V processor - HPET for precise timing - GPIO for sensors/actuators - SMBus for temperature/voltage monitoring - PM ACPI for power optimization - UART for debug/communication

### 13.4.2.3 3. Education Platform

Learn computer architecture: - Study peripheral internals - Practice interrupt handling - Understand power management - Experiment with different configurations - Debug with real hardware

### 13.4.2.4 4. SoC Prototyping

Rapid prototyping: - Test peripheral configurations - Validate interrupt routing - Optimize power consumption - Benchmark performance - Hardware/software co-development

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## 13.5 Nexys A7 Specific Integration

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### 13.5.1 Available Resources on Nexys A7

**Peripherals Already on Board:** - 16 switches (GPIO inputs) - 16 LEDs (GPIO outputs) - 5 push buttons - 7-segment displays (4 digits) - USB-UART bridge - PMOD connectors - VGA output - Microphone - Speaker - Temperature sensor - Accelerometer

### 13.5.2 How RLB Modules Map to Nexys A7

| RLB Module | Nexys A7 Resource       | Connection               |
|------------|-------------------------|--------------------------|
| GPIO       | Switches, LEDs, Buttons | Direct mapping           |
| UART_16550 | USB-UART bridge         | FT2232HQ chip            |
| HPET       | Internal timing         | No external pins needed  |
| PIT_8254   | Internal timing         | Timer/speaker            |
| RTC        | Internal clock          | Optional ext crystal     |
| PIC/IOAPIC | Internal routing        | IRQ aggregation          |
| SMBus      | I2C PMOD                | Via I2C PMOD adapter     |
| PM ACPI    | Power control           | Clock gates, power logic |

### 13.5.3 Example Nexys A7 Top-Level

```
module nexys_a7_rlb_system (  
    // Clock and reset  
    input wire      clk_100mhz,    // 100 MHz oscillator  
    input wire      cpu_resetsn,   // Reset button  
  
    // GPIO  
    input wire [15:0] sw,          // Switches → GPIO inputs  
    output wire [15:0] led,        // LEDs ← GPIO outputs  
    input wire [4:0] btn,          // Buttons → GPIO/interrupts  
  
    // UART  
    input wire      uart_rxd,      // USB-UART RX  
    output wire     uart_txd,      // USB-UART TX  
  
    // I2C/SMBus (via PMOD)  
    inout wire      smbus_sda,  
    inout wire      smbus_scl,  
  
    // 7-segment display  
    output wire [6:0] seg,  
    output wire [7:0] an,  
  
    // VGA (optional)  
    output wire [3:0] vga_r, vga_g, vga_b,  
    output wire      vga_hs, vga_vs  
);  
  
    // Instantiate soft processor (MicroBlaze, RISC-V, etc.)  
    // Instantiate APB crossbar  
    // Instantiate all RLB modules  
    // Connect to board resources  
  
endmodule
```

---

## 13.6 Implementation Recommendations

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### 13.6.1 For Nexys A7-100T

#### Recommended RLB Configuration:

|          |                                  |
|----------|----------------------------------|
| HPET     | - System timing                  |
| PIT_8254 | - Legacy timer for compatibility |
| RTC      | - Real-time clock                |

PIC\_8259 - Simple interrupt controller (OR IOAPIC)  
GPIO - Board I/O (switches, LEDs, buttons)  
UART\_16550 - Serial communication  
SMBus - I2C peripherals  
PM ACPI - Optional (if power features needed)

**Resource Budget:** - RLB modules: ~6,000 LUTs, ~5,000 FFs - MicroBlaze: ~2,000-3,000 LUTs - DDR2 Controller: ~2,000 LUTs - VGA/Display: ~1,000 LUTs - **Total:** ~11,000-12,000 LUTs (11% of XC7A100T)

**Plenty of room for:** - User applications - Additional peripherals - Debug logic - Custom IP cores

### 13.6.2 For Genesys 2

**Can Support Everything:** - All RLB modules simultaneously - Multiple soft processors - Complex memory hierarchy - High-speed peripherals - Video processing - Network interfaces

**Recommended for:** - Professional SoC development - Multi-core systems - High-performance applications - Complex retro systems (full PC replica)

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## 13.7 FPGA-Specific Considerations

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### 13.7.1 Clock Management

**Nexys A7:** - 100 MHz main clock → Can generate all needed frequencies - Use MMCM/PLL for: - CPU clock (50-200 MHz) - Peripheral clocks (25-100 MHz) - Display clocks (25.175 MHz for VGA) - USB clock (60 MHz)

**RLB Modules:** Most work at system clock (50-100 MHz), some support CDC for different clocks

### 13.7.2 Power Management

**PM ACPI Use on FPGA:** - Clock gating controls → Xilinx clock enables - Power domains → Can't control FPGA power, but can simulate - Useful for: Power-aware designs, testing, learning ACPI - Can control external power via GPIO pins

### 13.7.3 Interrupt Routing

**IOAPIC vs PIC\_8259 Choice:** - **Simple systems (single CPU):** Use PIC\_8259 (smaller) - **Advanced systems:** Use IOAPIC (more flexible, PC-compatible) - **Maximum flexibility:** Include both, use jumpers/config

### 13.7.4 External Interfaces

**How to Connect RLB to Board Resources:**

#### 1. GPIO Module:

- Inputs: Switches, buttons
  - Outputs: LEDs, 7-segment displays
  - Bidirectional: PMOD I/O pins
2. **UART\_16550:**
- Connect to FT2232HQ USB-UART
  - Baud rate generator uses system clock
  - CTS/RTS for flow control (optional)
3. **SMBus:**
- Connect to I2C PMOD
  - On-board: Temperature sensor (ADT7420)
  - On-board: Accelerometer (ADXL362)
  - External: I2C EEPROMs, sensors via PMOD
4. **Timers (HPET/PIT):**
- Internal only, no external pins
  - Can drive speaker output
  - Can generate PWM via GPIO
5. **RTC:**
- Internal timekeeping
  - Optional: External 32.768 kHz crystal
  - Battery backup simulation
- 

## 13.8 Soft Processor Options

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### 13.8.1 MicroBlaze (Xilinx)

- **Pros:** Well-supported, debugger, IP integration
- **Cons:** Proprietary, Vivado-specific
- **RLB Compatibility:** Excellent (APB via AXI interconnect)

### 13.8.2 RISC-V (Open Source)

- **Options:** VexRiscv, PicoRV32, Rocket, BOOM
- **Pros:** Open-source, portable, modern ISA
- **Cons:** Need to build toolchain
- **RLB Compatibility:** Excellent (APB native or via AXI)

### 13.8.3 Custom Soft Core

- Can design custom CPU specifically for RLB peripherals
  - Full control over instruction set
  - Educational value
- 

## 13.9 Step-by-Step FPGA Implementation

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### 13.9.1 Phase 1: Basic System (Week 1-2)

1. **Create FPGA project** (Vivado for Xilinx)
2. **Add simple processor** (MicroBlaze or RISC-V)
3. **Add APB crossbar** components/fabric-gen-ip/apbx-xbar/
4. **Add one RLB module** (start with GPIO)
5. **Connect to LEDs/switches**
6. **Test basic register access**

### 13.9.2 Phase 2: Expand Peripherals (Week 3-4)

7. **Add UART\_16550** → Test serial communication
8. **Add HPET** → Test precise timing
9. **Add PIC\_8259 or IOAPIC** → Test interrupts
10. **Add RTC** → Test time keeping

### 13.9.3 Phase 3: Advanced Features (Week 5-6)

11. **Add SMBus** → Test I2C communication
12. **Add PM\_ACPI** → Test power management
13. **Add all remaining RLB modules**
14. **System integration testing**

### 13.9.4 Phase 4: Application Development (Week 7+)

15. **Develop firmware** using RLB peripherals
  16. **Test real-world applications**
  17. **Optimize and debug**
  18. **Hardware/software co-verification**
-

## 13.10 Example Address Map for FPGA

| Base Address | Module     | Size  | Description                          |
|--------------|------------|-------|--------------------------------------|
| -----        | -----      | ----- | -----                                |
| 0xFEC0_0000  | HPET       | 4KB   | High Precision Timer                 |
| 0xFEC0_1000  | PIC_8259   | 4KB   | Interrupt Controller (master)        |
| 0xFEC0_2000  | PIT_8254   | 4KB   | Programmable Interval Timer          |
| 0xFEC0_3000  | RTC        | 4KB   | Real-Time Clock                      |
| 0xFEC0_4000  | SMBus      | 4KB   | System Management Bus                |
| 0xFEC0_5000  | PM ACPI    | 4KB   | Power Management                     |
| 0xFEC0_6000  | IOAPIC     | 4KB   | I/O APIC                             |
| 0xFEC0_7000  | GPIO       | 4KB   | General Purpose I/O                  |
| 0xFEC0_8000  | UART_16550 | 4KB   | Serial Port                          |
| 0xFEC0_9000  | PIC_8259   | 4KB   | Interrupt Controller (cascade slave) |

All accessible via APB from soft processor. The base is the BASE\_ADDR parameter of rlb\_top (default 0xFEC00000); the window order is the generated apbx\_xbar\_1to10's and is not free to rearrange.

## 13.11 Software Development

### 13.11.1 Bare-Metal Firmware

Using RLB Modules in C:

```
#include "rlb_peripherals.h"

// Initialize system
void system_init(void) {
    // Configure GPIO
    gpio_set_direction(0xFFFF0000); // Upper 16 = outputs

    // Initialize UART
    uart_init(115200);
    uart_puts("System starting...\n");

    // Configure HPET timer
    hpet_init();
    hpet_start_timer(0, 1000000); // 1ms periodic

    // Setup interrupts via IOAPIC
    ioapic_configure_irq(14, 0x2E, IRQ_EDGE, 0); // IDE
    ioapic_configure_irq(15, 0x2F, IRQ_EDGE, 0); // IDE
```

```

    // Enable power management
    pm_acpi_init();
}

```

### 13.11.2 Operating System Support

**Can Run:** - **Bare-metal:** Direct hardware access - **RTOS:** FreeRTOS, Zephyr with RLB drivers - **Linux:** With proper device drivers for RLB - **DOS/DOS-compatible:** If using x86-compatible core - **Custom OS:** Educational OS development

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## 13.12 Advantages of RLB on FPGA

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### 13.12.1 Development Benefits

1. **Rapid Prototyping:** Change peripheral config instantly (no PCB)
2. **Debugging:** ChipScope/ILA for internal signals
3. **Flexibility:** Easy to add/remove peripherals
4. **Cost-Effective:** One board, infinite configurations
5. **Educational:** See everything, change everything

### 13.12.2 Technical Benefits

1. **Known-Good IP:** Well-documented, tested peripherals
2. **Standard Interfaces:** APB everywhere, easy integration
3. **Scalable:** Start simple, add complexity
4. **Portable:** Pure SystemVerilog, vendor-independent
5. **Modern Design:** Clean code, proper methodology

### 13.12.3 Vs. Microcontroller

Why FPGA + RLB instead of MCU?

| Feature        | MCU           | FPGA + RLB            |
|----------------|---------------|-----------------------|
| Peripherals    | Fixed         | Infinite flexibility  |
| Interrupts     | Fixed routing | Programmable (IOAPIC) |
| Timers         | Limited count | As many as needed     |
| Clock domains  | 1-2           | Unlimited with CDC    |
| Custom logic   | Limited       | Full flexibility      |
| Learning value | Black box     | Full visibility       |

| Feature | MCU   | FPGA + RLB              |
|---------|-------|-------------------------|
| Cost    | Lower | Higher but more capable |

## 13.13 Conclusion

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### 13.13.1 Can RLB Be Used on Nexys A7/Genesys2? ABSOLUTELY!

**Perfect Match Because:** - Plenty of logic resources - Rich I/O for peripheral connections - Clock resources for multi-domain design - Development boards with extensive support - Educational and professional use

**Recommended Starting Point:** 1. Nexys A7-100T for most projects 2. Genesys 2 for complex/multi-core systems 3. Start with GPIO + UART + HPET 4. Add peripherals incrementally 5. Full RLB system fits comfortably with room to spare

**The RLB modules are specifically designed to be FPGA-friendly and work excellently on these development boards!**

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**Last Updated:** 2025-11-16 **Recommendation:** Nexys A7-100T is ideal for complete RLB-based SoC development